

Teacher Supervision over Representation Equivalence Classes:

Why Distillation Transfers Capability Through the Output Function, Not the
Features

Sang-Il Han^{1,2}

¹Korea University of Technology and Education ²Biomonster Co.

sihan@koreatech.ac.kr

April 2026

Abstract

Knowledge distillation is usually framed as a choice of *what* to match in the teacher—its logits, hidden features, or sample relations—which presupposes that the teacher’s representation has absolute coordinates to match. It does not: a pretrained representation is identifiable only up to an orthogonal-and-isotropic-scaling equivalence class, so a student should learn the teacher’s *equivalence class*, not its features. The organizing fact is that capability is the teacher’s *output function*, a class invariant that factors through the quotient by the class action—so an objective recovers capability exactly when it is defined there. This makes absolute feature matching ill-posed and admissible supervision a matter of targeting class invariants (Gram structure, CKA, principal subspaces) or aligning coordinates first, unifying feature matching, relational distillation, alignment, and grafting in one geometric account. We validate on Qwen2.5 and Llama-3.1: a restoration study recovers a corrupted model’s representation (CKA ≈ 0.99) but *not* its capability, and an ablation isolates the cause—output-function (logit) matching drives capability while matching hidden representations aligns geometry without restoring function. Recovery is confined to the corpus-covered region, and a graft study confirms boundary overlap predicts transplant success but is necessary, not sufficient.

1 Introduction

Here is a puzzle. Take a corrupted language model and restore its internal representation until it almost exactly matches the teacher’s—per-layer CKA ≈ 0.99 , by a label-free signal. The model stays broken: its perplexity is 10^2 – $10^4\times$ worse and its top-1 agreement with the teacher is near zero. Now instead match the teacher’s *output distribution* and ignore the internal representation: capability returns (perplexity ratio ≈ 1.01 , top-1 0.98), even though the representation is no better aligned than before. Matching what is inside the network does not transfer what the network can do; matching what comes out of it does. Why? This paper’s answer is geometric, and it is the reason for the machinery that follows: a representation has no absolute coordinates, so “match the features” is the wrong target—capability is an *invariant* of the representation’s equivalence class, namely its output function, and only an objective defined on that invariant transfers it.

Before any of that machinery, it helps to fix what a model *is*. A network maps an input x to an output $f(x) = W(H(x))$: the input is carried to an internal representation $H(x)$, and a reader W (the downstream weights) turns that representation into the output. Because the reader can absorb any rotation or rescaling of the representation, the same function f is computed by a whole family of pairs (H, W) , so a model is properly not a single (H, W) but the *class* of all pairs computing the same output. Capability is a property of that class, not of H alone—which is exactly why matching H can leave the function untouched. The underlying fact fits in a few lines of code. A representation is just an intermediate activation, and the layer that reads it can absorb a change of its coordinate frame:

```

h  = x @ W1          # a hidden representation
y  = h @ W2          # the layer that reads it -> output

Q  = random_rotation() # any orthogonal change of basis
h2 = h @ Q            # rewrite the representation...
W2 = Q.T @ W2         # ...and the reader, together

assert allclose(y, h2 @ W2) # same output, always

```

The rotated network stores completely different numbers in `h2` yet computes the identical function. So the specific coordinates of `h` carry no information the model can use: they are an arbitrary frame, fixed only up to the rotations (and rescalings) the next layer undoes. This is the whole premise in miniature—a *feature has coordinates, the coordinates have no meaning, and only the function is determined*—and it is exactly why matching a teacher’s features basis-for-basis differs from matching its output, which is invariant to the rewrite above. The rest of the paper turns this three-line `assert` into a statement about equivalence classes and the quotients they form; appendix G gives the same three objects this induces—orbit, fiber, quotient—as runnable code for the reader who prefers to meet them that way first.

Two words in that sentence carry the whole paper, and both have exact analogues a systems or compiler reader already owns. A *feature* (the vector `h` above) is an *internal representation used by a computation*—the neural-network analogue of a compiler’s intermediate representation (IR). Just as renaming an SSA temporary changes the IR while computing the same program, $h \mapsto h @ Q$ changes the feature while computing the same function: the coordinates of an IR, or of a feature, are an implementation detail, not meaning. An *invariant* is the dual notion—the quantity that does *not* change under such a rewrite. This paper’s one empirical claim, in that vocabulary, is simple: the invariant a downstream consumer can rely on is the *output function*, not the feature. The feature is the model’s IR, free to be recoordinated; capability is the invariant.

Distillation and module transfer both raise the same question: in what coordinate system does the “knowledge” of a representation live? The established lines of distillation each answer a *what-to-match* question: logit distillation [11] matches output distributions, feature distillation [20] matches hidden activations, and relational distillation [17, 22] matches sample-to-sample structure. All three presuppose that there is a well-defined teacher quantity to match. We step back one level and question that premise. A student should reproduce the teacher’s features basis-for-basis only if those features have an absolute basis to begin with—and they do not: representations are unidentifiable up to a group of transformations. The consequence is not a new loss but a change in what teacher supervision *is*. The teacher is not a set of feature vectors but an equivalence class of them, and distillation is the transfer of that class (made precise in definition 5.1). We make this precise: absolute-coordinate matching is ill-posed, and transfer must target the *invariants* of the representation’s equivalence class, or operate in coordinates fixed by an explicit alignment. Read this way, feature matching, relational distillation, alignment-based transfer, and module grafting are not separate techniques but different points in one geometric formulation, distinguished by which part of the class structure they fix.

The distinction from prior work is sharp, and worth stating before a reader files this under “another use of CKA.” The difference is one of *question*, not method. Representation-similarity research [12, 13] asks *how alike* two representations are, and answers with a diagnostic number. We ask a different question: *what transfers a capability?* We therefore propose no new similarity measure and no new loss. We change the *object* of distillation: the representation is itself an equivalence class, and “what should we match” becomes “what is defined on the quotient the output function factors through.”

This reframing places CKA precisely. It is one coordinate on the representation quotient—the *coarser* of the two, which forgets the reader (section 3). Capability instead lives on the finer, joint quotient (proposition 3.2), so CKA is the *wrong* coordinate for transferring it. A similarity paper would stop at measuring it; our point is the opposite. A high CKA can coexist with zero capability transfer (table 3), precisely because the two live on different quotients.

Our contribution is a single conceptual reframing of teacher supervision—from matching teacher *features* to transferring the teacher’s representation *equivalence class*—in three parts. **Theory:** absolute-coordinate feature matching is ill-posed (section 2), and the output function is the class invariant capability depends on, factoring through the quotient as $f = \bar{f} \circ \pi_{\text{joint}}$ (proposition 3.2). **Loss taxonomy:** this factorization sorts basis-invariant objectives and locates standard logit and feature distillation within it (section 5), and predicts graft success from subspace overlap (section 6). **Experiment:** restoration, ablation, and graft studies on Qwen2.5 and Llama-3.1 validate the account (sections 6 and 7).

These are not three separate stories but one: teacher supervision transfers an equivalence class, and its operational core is a single distinction—a representation is an equivalence class, but capability is the output function, a class invariant (proposition 3.2).

Terms, fixed once. We use four words in fixed senses. The *representation* is the layer activations $\mathbf{H}$, defined only up to the class $[\mathbf{H}]$ (definition 2.1); the *output function* f maps input to logits and is a joint-class invariant (proposition 3.2); *capability* is any scalar read off that function (perplexity, top-1)—deliberately narrow, *not* a claim about reasoning or generalization beyond what the logits encode; and a teacher’s *knowledge*, informally, means exactly that output function, the transferable content. The thesis relates them: capability rides on the output function, invariant to the representation’s coordinates. The narrowness has teeth—one-step functionals can be restored while trajectory behavior is not (section 7.8)—so “capability transfers” always means this output-function sense and no more.

Each experiment tests a consequence of this one statement. The restoration study (section 7) shows the two come apart—aligning the representation (CKA ≈ 1) does not restore the function; restoring the function (output-function matching) does not require aligning the representation. The decoder-from-scratch result shows what follows when only the class-invariant output function is targeted: the teacher is rebuilt on whatever subspace the corpus covers. The graft study shows the same geometry governs a different operation—module transfer succeeds to the extent the boundary subspaces overlap. Read together, they say that the teacher’s transferable content is its output function (a class invariant), and “what transfers” is the part of that function one’s objective and corpus actually target.

2 Representational Unidentifiability

Fix a finite *probe set* $\mathcal{X} = \{x_1, \dots, x_N\}$ of inputs, and let $\mathbf{H} = \mathbf{H}(\mathcal{X}) \in \mathbb{R}^{N \times d}$ collect the d -dimensional representations of a chosen layer, one row $h_\ell(x_i)$ per probe. All representational quantities below— $\mathbf{H}$, its Gram matrix, CKA, the equivalence class $[\mathbf{H}]$, and the quotients of section 3—are defined relative to $\mathcal{X}$: they describe what the representation encodes *about this probe set*. We keep $\mathcal{X}$ fixed throughout a given comparison and suppress it in the notation when no ambiguity arises, writing $\mathbf{H}$ for $\mathbf{H}(\mathcal{X})$; where the choice of probes matters (the corpus-coverage results of section 7) we make the dependence explicit. The pointwise object $h_\ell(x) \in \mathbb{R}^d$ of proposition 3.1 is the single-input representation; the probe matrix $\mathbf{H}(\mathcal{X})$ stacks it over $\mathcal{X}$. Throughout, $O(d) = \{\mathbf{Q} \in \mathbb{R}^{d \times d} : \mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}\}$ denotes the orthogonal group in dimension d (rotations

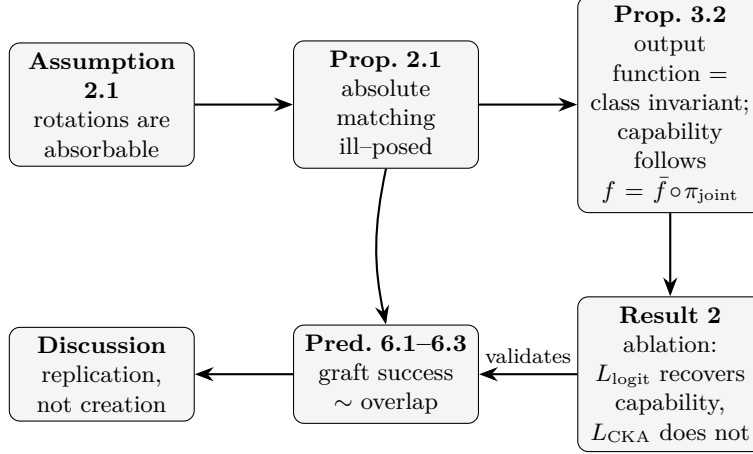


Figure 1: The logical spine. assumption 2.1 (rotations can be absorbed) makes absolute feature matching ill-posed (proposition 2.1) and identifies the output function as the class invariant capability depends on (proposition 3.2, the central result). The ablation of Result 2 (section 7.3) is its experimental counterpart; the same geometry yields falsifiable graft predictions (Predictions 6.1–6.3), validated in section 6; the Discussion places the whole as *replication*, not creation. Every later section is a consequence of the single premise on the left.

and reflections; $\mathbf{Q}^{-1} = \mathbf{Q}^\top$), acting on representations by right multiplication $h \mapsto h\mathbf{Q}$.

Definition 2.1 (Representation). Fix a probe set $\mathcal{X} = \{x_1, \dots, x_N\}$ and a layer ℓ of the network. The *representation* at that layer is the matrix

$$\mathbf{H} = \mathbf{H}(\mathcal{X}) \in \mathbb{R}^{N \times d}, \quad \text{row } i \text{ of } \mathbf{H} = h_\ell(x_i) \in \mathbb{R}^d, \quad (1)$$

whose i th row is the layer- ℓ hidden activation elicited by probe x_i (the residual-stream state after block ℓ , in a transformer). It is a purely empirical object: the network’s internal features *sampled* on $\mathcal{X}$, and it is defined only relative to $\mathcal{X}$ —a different probe set yields a different $\mathbf{H}$. The group $\mathcal{G} = \text{O}(d) \times \mathbb{R}_+$ acts on it by $\mathbf{H} \mapsto c\mathbf{H}\mathbf{Q}$; this action is what definition 2.2 quotients to form the equivalence class $[\mathbf{H}]$.

Naming, fixed once. Throughout, $\mathbf{H}$ (the layer activations) is the *representation* and $\mathbf{W}$ (specifically $\mathbf{W}_{\text{in}}^{(\ell)}$) is the *reader*; the function depends on the pair together, so neither alone is “the representation” (appendix E defends this usage against the representation-theory and computer-science readers’ objections).

A released model fixes θ , not a representation $\mathbf{H}$: the shipped weights commit to one coordinate frame of the joint class, which is why coordinate-sensitive operations across independently released models need an explicit alignment (definition 4.1). Appendix C makes this precise.

Intuitively, let $\mathbf{W}_{\text{in}}^{(\ell)}$ denote the first downstream operator that consumes the representation h_ℓ , that is, the linear map carrying one layer to the next,

$$h_\ell \xrightarrow{\mathbf{W}_{\text{in}}^{(\ell)}} h_{\ell+1}.$$

Assumption 2.1 states that any admissible coordinate change of h_ℓ can be absorbed by a counter-reparametrization of $\mathbf{W}_{\text{in}}^{(\ell)}$, leaving the network function unchanged—rotate the features one way, rotate the weights that read them the other way, and the two cancel. (We write $\mathbf{W}_{\text{in}}$ for $\mathbf{W}_{\text{in}}^{(\ell)}$ when

the layer is fixed.) This is the single premise from which the rest of the paper follows. Appendix A discusses when it holds exactly (a plain MLP) and when only approximately (transformers, where normalization can break it).

Assumption 2.1 (Absorbable reparametrization). Let the map from the layer- ℓ representation to the output factor as $\phi_{\ell:L} = \rho \circ \psi : \mathbb{R}^d \rightarrow \mathbb{R}^V$, where the first stage consumes the representation through a linear operator $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d \times m}$ followed by a nonlinearity, $\psi(h) = \tilde{\psi}(h \mathbf{W}_{\text{in}})$, and any normalization between h and $\mathbf{W}_{\text{in}}$ commutes with $\mathbf{Q}$ (exactly on the subgroup of appendix A). Then for $\mathbf{Q} \in \text{O}(d)$ and $c \in \mathbb{R}_+$ the paired actions $h \mapsto ch\mathbf{Q}$ and $\mathbf{W}_{\text{in}} \mapsto c^{-1}\mathbf{Q}^\top \mathbf{W}_{\text{in}}$ leave the consumed product invariant, giving the *absorbing identity*

$$\underbrace{\tilde{\psi}((h\mathbf{Q})(\mathbf{Q}^\top \mathbf{W}_{\text{in}})) = \tilde{\psi}(h\mathbf{W}_{\text{in}})}_{\text{(2a) cancellation: } \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}} \implies \underbrace{\phi_{\ell:L}^{\mathbf{Q}}(h\mathbf{Q}) = \phi_{\ell:L}(h)}_{\text{(2b) function unchanged}}, \quad (2)$$

so the full group acting absorbably is $\mathcal{G} = \text{O}(d) \times \mathbb{R}_+$.

A caveat on the group. The full $\text{O}(d)$ action is exact in the idealized linear/MLP case; in a real transformer, where attention $Q/K/V$ projections, MLP projections, the residual stream, and RMSNorm gains are coupled, the exact group is the *architecture-preserving subgroup* of reparametrizations that every downstream operator can absorb, with the full $\text{O}(d)$ serving as the organizing approximation. Everything below is stated for $\text{O}(d)$ for clarity; the claims that matter—ill-posedness of absolute matching, invariance of the output function—hold verbatim on the subgroup, since it is a subgroup, and the appendix MLP (table 8) exhibits the exact case.

Here $\phi_{\ell:L}^{\mathbf{Q}}$ denotes $\phi_{\ell:L}$ with $\mathbf{W}_{\text{in}}$ so reparametrized; the cancellation is $(ch\mathbf{Q})(c^{-1}\mathbf{Q}^\top \mathbf{W}_{\text{in}}) = h\mathbf{W}_{\text{in}}$, the two actions exactly inverse on the product $h\mathbf{W}_{\text{in}}$ that every downstream computation sees. We state eq. (2) for $c = 1$ to keep the rotation visible; the scale factor reappears in propositions 3.1 and 3.2. Conventions: $h \in \mathbb{R}^{1 \times d}$ is one row $h_\ell(x)$ of the probe matrix, so the assumption is pointwise in x and the statement for $\mathbf{H}(\mathcal{X})$ follows by stacking over rows, using row vectors throughout.

Appendix B instantiates $\mathbf{W}_{\text{in}}$, ψ , and ρ concretely for a transformer (the residual stream and its query/key/value projections) and for a plain MLP, where the absorbing identity is exact.

Exact vs. approximate. The absorbing identity is exact when the normalization between $\mathbf{H}$ and $\mathbf{W}_{\text{in}}$ commutes with $\mathbf{Q}$ —plain RMSNorm does; a learned per-channel gain γ does so only on the subgroup it preserves, and is otherwise approximate with error set by the spread of γ (appendix A).

Status of the propositions. On the subgroup $\mathcal{G}_0$ where assumption 2.1 holds exactly, propositions 2.1, 3.1 and 3.2 are *exact theorems*; only the *size* of the equivalence class is approximate for general $\mathbf{Q}$, so the qualitative claims hold as soon as $\mathcal{G}_0$ is nontrivial, which it always is (appendix A).

The remainder of the paper develops the consequences of assumption 2.1, one per section. Once some non-identity $\mathbf{Q}$ preserves the function, the representation is identified only up to $\mathbf{Q}$: it is an equivalence class, not a fixed matrix (definition 2.2). Matching absolute coordinates is then ill-posed (proposition 2.1), so only class invariants are admissible targets (section 3). The output function is invariant under the joint action, hence a function on the quotient (propositions 3.1 and 3.2). And logit distillation is singled out as the objective already defined there (section 5).

Definition 2.2 (Representation equivalence class). For $\mathbf{H} \in \mathbb{R}^{N \times d}$ define

$$[\mathbf{H}] = \{ c\mathbf{H}\mathbf{Q} : \mathbf{Q} \in \text{O}(d), c \in \mathbb{R}_+ \}, \quad (3)$$

the orbit of $\mathbf{H}$ under right multiplication by orthogonal matrices and *isotropic* positive scaling. We restrict to isotropic scaling deliberately: the invariants we use below (Gram structure, CKA)

are invariant to $O(d)$ and to isotropic scaling, but *not* to a non-isotropic diagonal Λ , since $\mathbf{H}\Lambda$ changes $\mathbf{H}\mathbf{H}^\top$. The relevant identifiability group for these invariants is therefore $O(d) \times \mathbb{R}_+$, and we define $[\mathbf{H}]$ to match it exactly.

Student and teacher representations. The distillation setting compares two representations of the *same* probe set $\mathcal{X}$: the teacher’s $\mathbf{H}_T = \mathbf{H}_T(\mathcal{X}) \in \mathbb{R}^{N \times d_T}$ and the student’s $\mathbf{H}_S = \mathbf{H}_S(\mathcal{X}) \in \mathbb{R}^{N \times d_S}$, each a chosen layer’s representation evaluated on the common probes. Two conditions are needed for an absolute feature-matching loss $\|\mathbf{H}_T - \mathbf{H}_S\|_F$ to be even well-posed as a formula, and they are worth stating because they are exactly what the equivalence-class view problematizes. First, the two must be evaluated on the *same* $\mathcal{X}$, so the rows correspond (a representation is only defined relative to its probe set, section 2); we assume this throughout. Second, they must share dimension, $d_S = d_T = d$, so that the difference lives in a single space $\mathbb{R}^{N \times d}$ and an orthogonal $\mathbf{Q} \in O(d)$ can act on either. We take $d_S = d_T = d$ for proposition 2.1 and the relational constructions of section 3; the unequal-dimension case $d_S \neq d_T$, where no common $\mathbb{R}^{N \times d}$ exists and a learned map must bridge the two, is exactly what the alignment of section 4 (remark 4.1) handles. Even with both conditions met, the next proposition shows the loss is still ill-posed—not for lack of a shared space, but because that space carries no absolute basis.

Proposition 2.1 (Ill-posedness of absolute feature matching). Let $\mathbf{H}_T, \mathbf{H}_S \in \mathbb{R}^{N \times d}$ be teacher and student representations of a common probe set $\mathcal{X}$ (same dimension d , as above), and consider the feature-matching loss $L_{\text{abs}}(\mathbf{H}_S) = \|\mathbf{H}_T - \mathbf{H}_S\|_F^2$. Under assumption 2.1, L_{abs} is not constant on the student equivalence class $[\mathbf{H}_S]$: there exist $\mathbf{Q} \in O(d)$ with $L_{\text{abs}}(\mathbf{H}_S \mathbf{Q}) \neq L_{\text{abs}}(\mathbf{H}_S)$ while $\mathbf{H}_S \mathbf{Q}$, once completed by the corresponding downstream reparametrization $\mathbf{W}_{\text{in}} \mapsto \mathbf{Q}^\top \mathbf{W}_{\text{in}}$ (assumption 2.1), induces the same network function as $\mathbf{H}_S$ with $\mathbf{W}_{\text{in}}$. Consequently L_{abs} assigns different values to representations that are functionally equivalent once their readers are accounted for, and its minimizer is an artifact of the arbitrary representative chosen for $[\mathbf{H}_S]$.

Proof. Functional invariance of $\mathbf{H}_S \mathbf{Q}$ is immediate from assumption 2.1. For non-constancy, expand

$$L_{\text{abs}}(\mathbf{H}_S \mathbf{Q}) = \|\mathbf{H}_T\|_F^2 + \|\mathbf{H}_S\|_F^2 - 2 \operatorname{tr}(\mathbf{Q}^\top \mathbf{H}_S^\top \mathbf{H}_T).$$

Only the last term depends on $\mathbf{Q}$. Let $\mathbf{H}_S^\top \mathbf{H}_T = \mathbf{U} \Sigma \mathbf{V}^\top$ be a singular value decomposition with $\Sigma \neq 0$ (generic teacher and student). The map $\mathbf{Q} \mapsto \operatorname{tr}(\mathbf{Q}^\top \mathbf{U} \Sigma \mathbf{V}^\top)$ ranges over $[-\sum_i \sigma_i, \sum_i \sigma_i]$ as $\mathbf{Q}$ varies over $O(d)$, taking its maximum at $\mathbf{Q} = \mathbf{U} \mathbf{V}^\top$ (so that $\mathbf{Q}^\top \mathbf{U} \Sigma \mathbf{V}^\top = \mathbf{V} \Sigma \mathbf{V}^\top \succeq 0$). Hence it is non-constant, so L_{abs} is non-constant on $[\mathbf{H}_S]$. $\square$

The proposition, measured. Proposition 2.1 can be turned into a direct measurement. Take a real teacher representation $\mathbf{H}$ (Qwen2.5-0.5B, layer 12, $d = 896$) and apply an orthogonal reparametrization $\mathbf{H} \mapsto \mathbf{H} \mathbf{Q}$, interpolating $\mathbf{Q}$ from the identity to a random rotation. The rotation is function-preserving in the sense of proposition 3.1: a compensating downstream reparametrization $\phi^{\mathbf{Q}}$ exists for which $f^{\mathbf{Q}} = f$. The table evaluates the distillation losses only on the two *representatives* $\mathbf{H}$ and $\mathbf{H} \mathbf{Q}$, which is exactly what a representational objective sees. We compare three losses on the pair $(\mathbf{H}, \mathbf{H} \mathbf{Q})$: absolute feature matching $\|\mathbf{H} - \mathbf{H} \mathbf{Q}\|_F^2$, the CKA loss $1 - \text{CKA}$, and the normalized-Gram loss L_{rel} . The two representatives are the *same information in rotated coordinates*, yet table 1 shows the feature-matching loss growing without bound (to $\sim 7 \times 10^5$ at a full rotation) while $1 - \text{CKA}$ and L_{rel} stay at 0 to machine precision for every rotation. Absolute matching penalizes a change that alters nothing about the function; the basis-invariant objectives are exactly blind to it. This is proposition 2.1 as an experimental fact; its *training-level* counterpart appears in section 7, where optimizing L_{cka} to $\text{CKA} \rightarrow 1$ still leaves the function destroyed (table 3)—table 1 is the static form, the ablation the dynamic one.

Table 1: Proposition 2.1, measured on Qwen2.5-0.5B (layer 12, $d = 896$). A function-preserving rotation $\mathbf{Q}$ is interpolated from identity ($t = 0$) to a random orthogonal matrix ($t = 1$). Absolute feature matching diverges; the basis-invariant losses are identically zero. The feature-matching column reports L_{abs} of proposition 2.1 normalized per probe ($/N$).

rotation t	$\ \mathbf{H} - \mathbf{H}\mathbf{Q}\ _F^2/N$	$1 - \text{CKA}$	L_{rel}
0.00	0	0	0
0.10	1.1×10^4	0	0
0.25	6.9×10^4	0	0
0.50	2.5×10^5	0	0
0.75	4.9×10^5	0	0
1.00	7.0×10^5	0	0

This also settles a natural worry: is “CKA ≈ 1 yet capability zero” an artifact of CKA specifically—a known insensitivity of CKA to small-eigenvalue directions—rather than a fact about representation matching? It is not. The failure is not a property of CKA as a *measure* but of basis invariance as a *class* of objective. Proposition 2.1 is proved for any loss that is constant on $[\mathbf{H}_S]$, and table 1 exhibits the same blindness for L_{rel} (a normalized-Gram, kernel-free objective) as for CKA: both stay at 0 under a full function-preserving rotation. Any measure fine enough to distinguish the two representatives would, by definition, no longer be basis-invariant—and would then penalize the very rotations that leave the function unchanged, i.e. behave like absolute matching and fail for the opposite reason. The gap between CKA $\rightarrow 1$ and capability is therefore not a resolution limit of one similarity index but the structural fact that the output-selecting representative is quotiented away by every basis-invariant objective.

3 Admissible Targets: Equivalence-Class Invariants

Before exhibiting specific quantities, we say precisely what it means for one to be an invariant.

Definition 3.1 (Class invariant). A map q defined on representations is a $(\mathcal{G})$ -invariant if it is constant on each equivalence class, i.e.

$$q(c\mathbf{H}\mathbf{Q}) = q(\mathbf{H}) \quad \text{for all } \mathbf{Q} \in \text{O}(d), c \in \mathbb{R}_+, \quad (4)$$

equivalently if q factors through the quotient onto $\mathbb{R}^{N \times d}/\mathcal{G}$ (the representation quotient defined in eq. (7) below) as a well-defined function of the class. A function of *two* representations is an invariant if it is constant under the action applied to either argument separately. An invariant is thus a well-defined function of the class $[\mathbf{H}]$, not of the representative; it is exactly the kind of quantity a distillation objective may legitimately target, since by proposition 2.1 anything not of this form depends on the arbitrary basis.

We now exhibit three invariants: the Gram structure, CKA, and the principal-angle spectrum.

The first invariant is relational. The (uncentered) *Gram matrix* is $\mathbf{K} = \mathbf{H}\mathbf{H}^\top \in \mathbb{R}^{N \times N}$, with centered version $\widehat{\mathbf{K}} = \mathbf{C}\mathbf{K}\mathbf{C}$ where $\mathbf{C} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ ($\mathbf{1} \in \mathbb{R}^N$ the all-ones vector, $\mathbf{I}$ the $N \times N$ identity). It records inter-sample geometry, and is orthogonally invariant:

Lemma 3.1 (Orthogonal invariance of the Gram structure). For all $\mathbf{Q} \in \text{O}(d)$, $(\mathbf{H}\mathbf{Q})(\mathbf{H}\mathbf{Q})^\top = \mathbf{H}\mathbf{H}^\top$. Thus $\mathbf{K}$ and $\widehat{\mathbf{K}}$ are invariant under the orthogonal part of $[\mathbf{H}]$.

Proof. $(\mathbf{H}\mathbf{Q})(\mathbf{H}\mathbf{Q})^\top = \mathbf{H}\mathbf{Q}\mathbf{Q}^\top\mathbf{H}^\top = \mathbf{H}\mathbf{H}^\top$ since $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$. $\square$

The Gram matrix records inter-sample geometry (distances and angles), which is exactly the information the representation carries about the probe set, stripped of the arbitrary coordinate frame.

Definition 3.2 (Centered Kernel Alignment). For representations $\mathbf{H}_A, \mathbf{H}_B$ with centered Gram matrices $\widehat{\mathbf{K}}_A, \widehat{\mathbf{K}}_B$, and writing $\langle A, B \rangle_F = \text{tr}(A^\top B)$ for the Frobenius inner product (with $\|A\|_F = \langle A, A \rangle_F^{1/2}$),

$$\text{CKA}(\mathbf{H}_A, \mathbf{H}_B) = \frac{\langle \widehat{\mathbf{K}}_A, \widehat{\mathbf{K}}_B \rangle_F}{\|\widehat{\mathbf{K}}_A\|_F \|\widehat{\mathbf{K}}_B\|_F} \in [0, 1]. \quad (5)$$

Linear CKA was introduced by Kornblith et al. [13] as a representation similarity index invariant to orthogonal transformation and isotropic scaling, building on kernel-target alignment [5] and centered alignment [4]; it is a normalized Hilbert–Schmidt independence criterion [9].

Corollary 3.1. CKA is invariant under orthogonal transformation and isotropic scaling of either argument, and therefore is a well-defined similarity on equivalence classes $[\mathbf{H}_A], [\mathbf{H}_B]$.

Proof. Orthogonal invariance follows from lemma 3.1. Isotropic scaling $\mathbf{H} \mapsto c\mathbf{H}$ scales $\widehat{\mathbf{K}}$ by c^2 , which cancels in the normalized ratio. $\square$

A third invariant measures shared subspace. If $\mathbf{H}_A, \mathbf{H}_B$ have orthonormal column bases $\mathbf{U}_A, \mathbf{U}_B$ and $k = \min(\text{rank } \mathbf{H}_A, \text{rank } \mathbf{H}_B)$, the *principal angles* $\theta_1 \leq \dots \leq \theta_k$ satisfy $\cos \theta_i = \sigma_i(\mathbf{U}_A^\top \mathbf{U}_B)$ (the i th singular value of $\mathbf{U}_A^\top \mathbf{U}_B$), and the *effective shared dimension* is $k_{\text{eff}} = \sum_{i=1}^k \cos^2 \theta_i$. The values $\{\cos^2 \theta_i\}$ quantify how much of the two feature dictionaries point in common directions; they will drive the graft predictions in section 6.

The output function is invariant under the joint class action. The invariants above (Gram, CKA, principal angles) are properties of the representation. We now make explicit the link the rest of the paper relies on, and to do so we name two distinct operations that the discussion repeatedly contrasts.

- A *rotation (alone)* acts on the representation only: $h \mapsto h\mathbf{Q}$ with the downstream parameters θ left unchanged. This *does* change the network function in general—feeding $h\mathbf{Q}$ into the unchanged downstream produces different logits (remark 3.1 below; empirically $\|f_{\text{rot}} - f\| \neq 0$, table 8).
- A *joint reparametrization* acts on the representation *and* the downstream together: for $(\mathbf{Q}, c) \in \mathcal{G} = \text{O}(d) \times \mathbb{R}_+$, $(h, \theta) \mapsto (ch\mathbf{Q}, \theta^{\mathbf{Q},c})$, where $\theta^{\mathbf{Q},c}$ is the absorbing change $\mathbf{W}_{\text{in}} \mapsto c^{-1}\mathbf{Q}^\top \mathbf{W}_{\text{in}}$ of assumption 2.1 (the rotation $\mathbf{Q}$ is undone by $\mathbf{Q}^\top$ and the scale c by c^{-1}). By eq. (2) this leaves the network function unchanged: $f^{\mathbf{Q},c} = f$.

The point of proposition 3.1 is that the output function is invariant under the *joint* reparametrization, and *not* under rotation alone. A reparametrization of the representation does not act on $\mathbf{H}_\ell$ in isolation: under assumption 2.1 the downstream blocks are reparametrized in compensation, and it is this joint action—on the representation *and* the downstream parameters—that the function is constant under. This is exactly why two networks differing by such an action implement the same function, and why capability transfers along the class; it is also why a representational objective, which sees only h and can move it by a rotation alone, is not by itself a capability objective.

Proposition 3.1 (Output invariance under the joint class action). Write the network function as $f(x) = \phi_{\ell:L}(h_\ell(x))$, with $\phi_{\ell:L} : \mathbb{R}^d \rightarrow \mathbb{R}^V$ the map from the layer- ℓ representation $h_\ell(x) \in \mathbb{R}^d$ to the output logits as in assumption 2.1 (it carries the remaining blocks and the unembedding). For $\mathbf{Q} \in \text{O}(d)$, let $\phi_{\ell:L}^{\mathbf{Q}}$ denote $\phi_{\ell:L}$ with its input operator $\mathbf{W}_{\text{in}}$ reparametrized to absorb $\mathbf{Q}$ (the existence of such a $\phi_{\ell:L}^{\mathbf{Q}}$ is assumption 2.1). Then for all x ,

$$\phi_{\ell:L}^{\mathbf{Q}}(h_\ell(x) \mathbf{Q}) = \phi_{\ell:L}(h_\ell(x)), \quad \text{hence} \quad f^{\mathbf{Q}}(x) = f(x), \quad (6)$$

where $f^{\mathbf{Q}}$ is the network with representation $h_\ell \mathbf{Q}$ and downstream $\phi_{\ell:L}^{\mathbf{Q}}$. Thus the pair (h_ℓ, θ) , with θ the downstream parameters, transforms under a joint equivalence action, and f is invariant under it: the output function—and any capability metric computed from the logits (perplexity, top-1 agreement)—is constant along the orbit $\{(ch_\ell \mathbf{Q}, \phi_{\ell:L}^{\mathbf{Q},c}) : (\mathbf{Q}, c) \in \mathcal{G}\}$. The invariant object is the orbit $[(h_\ell, \theta)]$; the pair (h_ℓ, θ) is what the action moves.

Proof. This is precisely eq. (2). The substitution $\mathbf{W}_{\text{in}} \mapsto \mathbf{Q}^\top \mathbf{W}_{\text{in}}$ gives the cancellation (2a), $\tilde{\psi}((h_\ell \mathbf{Q})(\mathbf{Q}^\top \mathbf{W}_{\text{in}})) = \tilde{\psi}(h_\ell \mathbf{W}_{\text{in}})$, so every downstream activation, and thus the logits, are unchanged; hence the consequence (2b), $\phi_{\ell:L}^{\mathbf{Q}}(h_\ell \mathbf{Q}) = \phi_{\ell:L}(h_\ell)$, and $f^{\mathbf{Q}} = f$. Invariance for all $\mathbf{Q} \in \text{O}(d)$ means f is constant on the joint orbit. $\square$

Remark 3.1. The invariance is *joint*: it is not true that feeding $h_\ell \mathbf{Q}$ into the *unchanged* $\phi_{\ell:L}$ preserves the output—that would generally change f . What is preserved is the function realized by the jointly-reparametrized network. Equivalently, the output function is a well-defined function on the quotient by the joint action, not on $[h_\ell]$ alone. This is the precise sense in which capability is a class invariant. An analogy: reaching the representation quotient is like walking into the right meeting room (the correct geometric structure); reaching the joint quotient is like sitting in the exact seat assigned to you (the coordinates the output head can read). CKA gets you into the room, but it does not tell you which seat is yours—which is why aligning the representation is not the same as restoring the function.

Remark 3.2 (Orbit and fiber, in programmers’ terms). Two words organize what follows, and both have exact programming analogues. An *orbit* is all implementations of one program: rewriting the two-layer network’s internals by any $(\mathbf{Q}, c) \mapsto ch_\ell \mathbf{Q}$ with the reader updated to $c^{-1} \mathbf{Q}^\top \mathbf{W}_2$ —gives different source code computing the same output, and the set of all such rewrites is the orbit of $(h, \mathbf{W}_2)$. A *fiber* is everything mapping to one behavior: for a map π and a fixed output y , the fiber $\pi^{-1}(y)$ collects all inputs π sends to y . Here π is the joint quotient map (defined next), and its fiber over a class is exactly one orbit—the clones computing the same function. The two words name the same subset; appendix G realizes both as runnable code. Everything that follows is bookkeeping for these sets: a legitimate distillation target may depend on *which* fiber the student lands in, never on *where inside* it sits.

The chain as a factorization through the quotient. Proposition 3.1 is best read through quotients by the class action, of which there are two, and distinguishing them is what makes the chain precise. Let $\mathcal{G} = \text{O}(d) \times \mathbb{R}_+$. It acts on representations alone by $h \mapsto ch\mathbf{Q}$. The *quotient space* $\mathbb{R}^{N \times d} / \mathcal{G}$ is the set of equivalence classes under this action, $\mathbb{R}^{N \times d} / \mathcal{G} = \{[h] : h \in \mathbb{R}^{N \times d}\}$ with $[h]$ as in eq. (3) (we use only the elementary theory of group actions and quotient spaces; see, e.g., 15); collapsing each orbit to a single point yields the *representation quotient map*

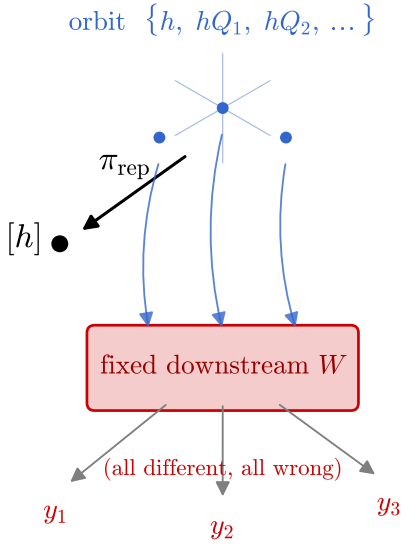
$$\pi_{\text{rep}} : \mathbb{R}^{N \times d} \longrightarrow \mathbb{R}^{N \times d} / \mathcal{G}, \quad \pi_{\text{rep}}(h) = [h], \quad (7)$$

on which the invariants of section 3 (CKA, normalized Gram, principal angles) are functions. But the network function does *not* live on $\mathbb{R}^{N \times d}/\mathcal{G}$: by remark 3.1 feeding $h\mathbf{Q}$ into the unchanged downstream changes f . The function lives on the *joint* quotient. Writing Θ for the downstream parameters and letting $\mathcal{G}$ act jointly, for $g = (\mathbf{Q}, c) \in \mathcal{G}$, $g \cdot (h, \theta) = (ch\mathbf{Q}, \theta^{\mathbf{Q},c})$ with $\theta^{\mathbf{Q},c}$ the absorbing reparametrization $\mathbf{W}_{\text{in}} \mapsto c^{-1}\mathbf{Q}^\top \mathbf{W}_{\text{in}}$ of assumption 2.1, define the *joint space* $\mathbf{M} = \mathbb{R}^{N \times d} \times \Theta$ and its quotient $\mathbf{M}/\mathcal{G} = \{[(h, \theta)] : (h, \theta) \in \mathbf{M}\}$, the set of joint orbits, with quotient map

$$\pi_{\text{joint}} : \mathbf{M} \longrightarrow \mathbf{M}/\mathcal{G}, \quad \pi_{\text{joint}}(h, \theta) = [(h, \theta)].$$

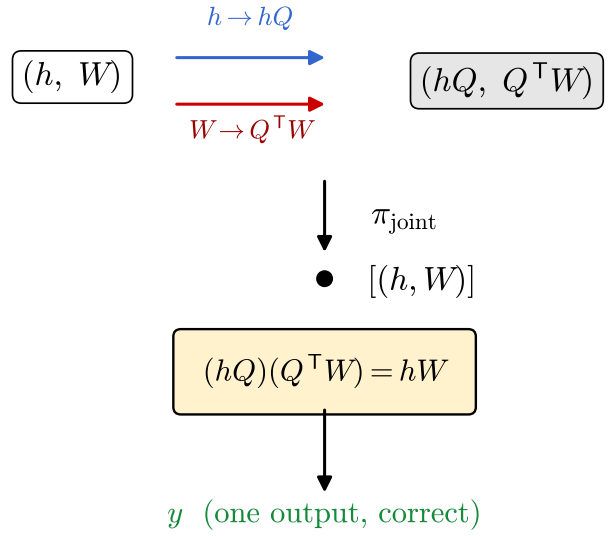
The invariants of section 3 live on π_{rep} ; the output function, as the next proposition shows, lives on π_{joint} .

Representation quotient $\mathbb{R}^{N \times d}/\mathcal{G}$
(collapse h only)



the same $[h]$ has many representatives;
a fixed W reads each in the wrong coordinates
 $\Rightarrow$ right room, wrong seat: capability lost

Joint quotient $\mathbf{M}/\mathcal{G}$
(collapse (h, W) together)



representation and reader move together;
the output is identical for every Q
 $\Rightarrow$ correct seat: $f^Q = f$, capability preserved

Figure 2: Why capability lives on the joint quotient, not the representation quotient (two-layer network of example 3.1). *Left*: collapsing h alone to $[h] \in \mathbb{R}^{N \times d}/\mathcal{G}$ is what CKA does—but if the downstream W is held fixed, the rotated representation is read in the wrong coordinates and the output is wrong, so matching $[h]$ does not restore capability. *Right*: the *joint* action rotates h and counter-rotates W together; the product $(h\mathbf{Q})(\mathbf{Q}^\top W) = hW$ is unchanged, so (h, W) and its variants collapse to one point $[(h, W)] \in \mathbf{M}/\mathcal{G}$ on which the output—and hence capability—is constant ($f^Q = f$, eq. (8)).

Example 3.1 (Both quotients on a two-layer network). Concretely, on $x \rightarrow h \rightarrow y$ with $y = h\mathbf{W}_2$: rotating h and counter-rotating the reader, $(h\mathbf{Q})(\mathbf{Q}^\top \mathbf{W}_2) = h\mathbf{W}_2 = y$ (and isotropic scale absorbed by $\mathbf{W}_2 \mapsto c^{-1}\mathbf{W}_2$), leaves the output fixed. So every $ch\mathbf{Q}$ is the same point $[h]$ of the representation

quotient (fig. 2, left)—but $[h]$ alone forgets which $\mathbf{W}_2$ reads it (remark 3.1). The pair $(h, \mathbf{W}_2)$ and its absorbable variants form one point of the *joint* quotient (fig. 2, right), and every network there computes the same function. This is eq. (8) below.

Proposition 3.2 (Factorization through the joint quotient). Intuitively, two networks differing only by an absorbable coordinate change are treated as the same point, and the function reads only that point. Formally, under assumption 2.1 the output function descends to the joint quotient: there is a map $\bar{f}$ on $\mathbf{M}/\mathcal{G}$ with

$$f = \bar{f} \circ \pi_{\text{joint}}, \quad \text{i.e.} \quad f \text{ depends on } (h_\ell, \theta) \text{ only through } [(h_\ell, \theta)]. \quad (8)$$

Consequently any evaluation functional κ computed from the logits (perplexity, top-1 agreement, KL to a reference) factors as $\kappa = \bar{\kappa} \circ \pi_{\text{joint}}$: capability is constant on fibers of π_{joint} (a *fiber* is one orbit—all the clones that compute the same function in different coordinates), hence a well-defined function on the joint quotient. It is *not*, in general, a function on the representation quotient $\mathbb{R}^{N \times d}/\mathcal{G}$ alone—that is exactly the content of remark 3.1.

Proof. The joint action $g \cdot (h, \theta) = (c h \mathbf{Q}, \theta^{\mathbf{Q}, c})$ of $\mathcal{G} = \text{O}(d) \times \mathbb{R}_+$ partitions $\mathbf{M}$ into orbits, which are exactly the fibers of π_{joint} . By proposition 3.1, for every $g \in \mathcal{G}$ the output is unchanged, $f(g \cdot (h, \theta)) = f(h, \theta)$ (the absorbing identity (2), an algebraic equality), so f is constant on each fiber. A map constant on fibers descends uniquely through the surjection π_{joint} (the universal property of the quotient): setting $\bar{f}([(h, \theta)]) := f(h, \theta)$ is well-defined by fiber-constancy and is the only $\bar{f}$ with $\bar{f} \circ \pi_{\text{joint}} = f$, which is (8). Any evaluation functional κ of the logits is a function of f , so $\kappa = \bar{\kappa} \circ \pi_{\text{joint}}$ inherits the same factorization. Every step is an equality over the subgroup on which assumption 2.1 is exact (appendix A). $\square$

We read eq. (8) with the input held fixed: for each x , the pair $(h_\ell(x), \theta)$ determines the logits, and $\bar{f}$ sends the joint class $[(h_\ell(x), \theta)]$ to $f(x) \in \mathbb{R}^V$. The probe set $\mathcal{X}$ enters through the representation $h_\ell(\cdot)$, so $\bar{f}$ is a function of the class at each input rather than of a single global orbit; the invariance is over $(\mathbf{Q}, c) \in \mathcal{G}$, not over x .

The factorization is what makes the four operations of the paper commensurable: a distillation objective is *admissible* exactly when it factors through the appropriate quotient—constant on fibers, hence a function on the quotient. Proposition 2.1 is the statement that absolute feature matching does *not* factor through π_{rep} (it separates points of a single fiber), which is why its minimizer is fiber-dependent and meaningless. The invariants of section 3 are coordinates on the representation quotient $\mathbb{R}^{N \times d}/\mathcal{G}$; the alignment of section 4 (definition 4.1) is a choice of representative within a fiber; and—the point developed next—logit distillation lives on the joint quotient $\mathbf{M}/\mathcal{G}$ by construction.

Proposition 3.2 is the theoretical hinge of the paper, and it settles the loss taxonomy of section 5 in one stroke. Because capability is determined by $f = \bar{f} \circ \pi_{\text{joint}}$, a function on the joint quotient, an objective can transfer capability only if it targets the output-function invariant on that quotient—being defined on the quotient is necessary (a loss that separates points of one fiber penalizes function-preserving reparametrizations), though not sufficient (a constant loss is defined on the quotient yet transfers nothing). This immediately sorts the candidate losses into two kinds. A loss written on the raw representation $\mathbb{R}^{N \times d}$ is *not* on the quotient as written. To be admissible it must be *pushed down* onto it—made constant on fibers. That is precisely the normalization turning $\|\mathbf{K}_S - \mathbf{K}_T\|$ into the class-invariant L_{rel} and L_{CKA} . A loss written on the output distribution, by contrast, is *already* a function on the joint quotient: there is no fiber left to quotient out. This is why logit distillation needs no such engineering and is the most stable objective. Every distinction

in section 5 is a reading of this one fact: hidden-state losses must be lowered to the quotient, the logit loss is born there.

This is the bridge between the two halves of the paper. The representational invariants (CKA, Gram, angles) say *when two representations are the same class*; propositions 3.1 and 3.2 say the *output function is a function on the quotient*, not a property of a chosen representative. It is why proposition 2.1 (absolute matching is ill-posed) and the empirical result that output-function matching—not representational matching—restores capability (section 7) are two sides of one fact: capability lives on the quotient, so an objective can recover it only by targeting the output-function invariant there.

4 Alignment Across the Equivalence Class

To match coordinates (not just relations), one must first move the student to a representative of the teacher’s class. We make “alignment” precise as an operation on the equivalence-class structure of section 2, before giving the map that realizes it.

Definition 4.1 (Alignment). Let $\mathbf{H}_A$ and $\mathbf{H}_B$ be representations of a common probe set $\mathcal{X}$, with $[\mathbf{H}_B] = \{c\mathbf{H}_B\mathbf{Q} : \mathbf{Q} \in \text{O}(d), c \in \mathbb{R}_+\}$ its equivalence class (eq. (3)). An *alignment of $\mathbf{H}_B$ to $\mathbf{H}_A$* is the choice of the representative of $[\mathbf{H}_B]$ closest to $\mathbf{H}_A$, i.e. the element

$$\mathbf{H}_B^\star = \arg \min_{\mathbf{H}' \in [\mathbf{H}_B]} \|\mathbf{H}_A - \mathbf{H}'\|_F = c^\star \mathbf{H}_B \mathbf{Q}^\star, \quad (9)$$

together with the group element $(\mathbf{Q}^\star, c^\star) \in \mathcal{G}$ that realizes it. Equivalently, alignment is a section of the representation quotient map π_{rep} over the fiber $\pi_{\text{rep}}^{-1}([\mathbf{H}_B])$: it selects one point of the fiber (one coordinate frame) rather than collapsing the fiber to the class. It is thus dual to the invariants of section 3, which quotient the fiber out; alignment instead fixes a representative within it. The attained value $\rho = \|\mathbf{H}_A - \mathbf{H}_B^\star\|_F / \|\mathbf{H}_A\|_F$ is the *alignment residual*, a class invariant of the pair (constant under separate reparametrizations of $\mathbf{H}_A, \mathbf{H}_B$) that measures how far the two classes are from coinciding. When $d_A \neq d_B$ no single fiber contains both, and $\mathbf{Q}^\star$ is replaced by a learned low-capacity map $\mathbf{W}$ (remark 4.1) that plays the same role of selecting coordinates.

Proposition 4.1 solves eq. (9) in closed form for the orthogonal part of the group (the scale $c^\star$ is fixed separately by $c^\star = \langle \mathbf{H}_A, \mathbf{H}_B \mathbf{Q}^\star \rangle / \|\mathbf{H}_B\|_F^2$).

Proposition 4.1 (Orthogonal Procrustes alignment). For $d_A = d_B = d$, the orthogonal rotation $\mathbf{Q}^\star$ that aligns $\mathbf{H}_B$ to $\mathbf{H}_A$ most closely is given in closed form by (10): writing the singular value decomposition $\mathbf{H}_B^\top \mathbf{H}_A = U \Sigma V^\top$, the minimizer is $\mathbf{Q}^\star = UV^\top$.

$$\mathbf{Q}^\star = \arg \min_{\mathbf{Q} \in \text{O}(d)} \|\mathbf{H}_A - \mathbf{H}_B \mathbf{Q}\|_F = UV^\top, \quad \mathbf{H}_B^\top \mathbf{H}_A = U \Sigma V^\top. \quad (10)$$

The aligned residual $\rho = \|\mathbf{H}_A - \mathbf{H}_B \mathbf{Q}^\star\|_F / \|\mathbf{H}_A\|_F$ is invariant under separate orthogonal reparametrizations of $\mathbf{H}_A, \mathbf{H}_B$ and measures how far the two representations are from being the same information in different coordinates. This is the orthogonal Procrustes problem [21], used in representational similarity analysis [14] to compare neural population codes.

Proof. Minimizing $\|\mathbf{H}_A - \mathbf{H}_B \mathbf{Q}\|_F^2$ over $\text{O}(d)$ is equivalent to maximizing $\text{tr}(\mathbf{Q}^\top \mathbf{H}_B^\top \mathbf{H}_A)$, whose maximizer is $UV^\top$ by the von Neumann trace inequality (saturated when $\mathbf{Q}^\top U \Sigma V^\top$ is symmetric positive semidefinite). Reparametrizing $\mathbf{H}_A \mapsto \mathbf{H}_A R_A$, $\mathbf{H}_B \mapsto \mathbf{H}_B R_B$ replaces $\mathbf{Q}^\star$ by $R_B^\top \mathbf{Q}^\star R_A$ and leaves the residual norm unchanged, giving invariance of ρ . $\square$

Remark 4.1 (Unequal dimensions). Proposition 4.1 assumes teacher and student share a hidden dimension ($d_A = d_B$), so a rotation $\mathbf{Q} \in \text{O}(d)$ can align their coordinate frames directly. In practice the two often differ—e.g. a 4096-wide teacher and a 2048-wide student—and then no orthogonal $\mathbf{Q}$ even exists, since rotation is defined only within a single dimension. For $d_A \neq d_B$ we therefore replace $\mathbf{Q}$ by a learned linear map $\mathbf{W} \in \mathbb{R}^{d_B \times d_A}$ solving $\min_{\mathbf{W}} \|\mathbf{H}_A - \mathbf{H}_B \mathbf{W}\|_F$ (ridge-regularized), the linear case of feature distillation. The map should be *low capacity*: alignment is coordinate correction, not new learning. In other words, $\mathbf{W}$ plays the role of a coordinate converter, not a knowledge generator—its job is to express the student’s existing representation in the teacher’s coordinate system, not to synthesize structure the student does not have. A high-capacity $\mathbf{W}$ (a deep MLP, say) defeats this: it can manufacture the teacher’s structure from an unrelated student, after which one can no longer tell whether the two representations were genuinely the same class or were forced to match by the bridging network—the comparison stops measuring alignment and starts measuring the capacity of $\mathbf{W}$.

5 Basis-Invariant Distillation Objectives

Distillation estimates the student parameters θ_S ; these induce a representation $\mathbf{H}_S(\mathcal{X})$ and an output function $f_S = f(\cdot; \theta_S)$, and the pair $(\mathbf{H}_S, \theta_S)$ determines the student’s joint class. The question of this section is which *loss* to minimize in that estimation. Proposition 3.2 immediately induces the following taxonomy: since a loss is admissible only if it lives on a quotient, every candidate falls into three families by *which* quotient that is (table 2).

Family	Target	Defined on	Alignment?	Recovers capability?
(A) Relational ($L_{\text{rel}}, L_{\text{CKA}}$)	representation geometry (Gram)	representation quotient $\mathbb{R}^{N \times d} / \mathcal{G}$	no	no (structure only)
(B) Align-then-match	aligned absolute coordinates	a chosen representative	yes (Procrustes / $\mathbf{W}$)	conditional (on alignment)
(C) Logit KD (L_{logit})	output distribution	joint quotient $\mathbf{M} / \mathcal{G}$	none needed	yes (targets the invariant)

Table 2: The three basis-invariant families, by where each loss lives relative to the quotient. (A) and (B) act on the hidden representation and must be made basis-free by hand; (C) is already a function on the joint quotient, which is why it alone targets capability directly (proposition 3.2).

Only the third family targets capability without requiring an additional coordinate-selection or reader-compatibility assumption, because capability lives on the joint quotient (proposition 3.2); the other two live on the coarser representation quotient and, on their own, transfer representation geometry rather than the output function. This is not to say a feature objective *cannot* contribute to capability—family (B), if it also learns or fixes the downstream reader so that the aligned frame is the one the reader expects, can—only that doing so reintroduces exactly the coordinate-selection step the logit objective never needs. The three are exhaustive because there are only three things a basis-invariant loss can do about the arbitrary coordinate frame: discard it (A), fix it first (B), or never read it (C)—and only the last is born on the quotient where capability lives. The rest of this section develops each in turn, then states the definition of admissibility that makes “defined on the quotient” precise.

We use the notation already established: $\mathbf{H}_S, \mathbf{H}_T \in \mathbb{R}^{N \times d}$ are the student and teacher representations of a common probe set $\mathcal{X}$ (section 2); $\mathbf{K} = \mathbf{H}\mathbf{H}^\top$ is the Gram matrix and $\widehat{\mathbf{K}} = \mathbf{C}\mathbf{K}\mathbf{C}$ its centered form, with CKA the normalized similarity built from them (definition 3.2); $\mathbf{W}$ is the low-capacity alignment map of remark 4.1.

(A) Relational matching (alignment-free). The first option throws the basis away entirely and compares only what survives it—the geometry of the representation, i.e. the angles and distances between the N probe points, which the Gram matrix records. Two representations of the same class have the same geometry, so matching geometry needs no alignment. To be invariant to the full $O(d) \times \mathbb{R}_+$ class, the Gram matrices must be normalized before comparison:

$$L_{\text{rel}} = \left\| \frac{\widehat{\mathbf{K}}_S}{\|\widehat{\mathbf{K}}_S\|_F} - \frac{\widehat{\mathbf{K}}_T}{\|\widehat{\mathbf{K}}_T\|_F} \right\|_F^2 \quad \text{or} \quad L_{\text{CKA}} = 1 - \text{CKA}(\mathbf{H}_S, \mathbf{H}_T). \quad (11)$$

Both are invariant to orthogonal transformation *and* isotropic scaling. We emphasize that the *unnormalized* Gram loss $\|\widehat{\mathbf{K}}_S - \widehat{\mathbf{K}}_T\|_F^2$ is orthogonal-invariant but *not* scale-invariant—under $\mathbf{H} \mapsto c\mathbf{H}$ the Gram scales as c^2 , so it is not constant on $[\mathbf{H}]$ and would reintroduce a (milder) version of the ill-posedness of proposition 2.1. Normalization (or, equivalently, the cosine form L_{CKA}) is required. No alignment is estimated, and the two arguments need not share dimension (both Gram matrices are $N \times N$); this is the loss to reach for when teacher and student have different widths. The cost is that absolute position is not transferred, which matters when a downstream module consumes the representation coordinatewise.

(B) Align-then-match. The second option keeps the basis but fixes it first: estimate the rotation (or low-capacity map) that moves the student onto a representative of the teacher’s class, then match coordinates in that aligned frame. This is the right choice when absolute position *is* needed—when a module will read the features coordinatewise, as in grafting.

$$L_{\text{align}} = \|\mathbf{H}_S \mathbf{W} - \mathbf{H}_T\|_F^2, \quad \mathbf{W} \text{ low capacity (proposition 4.1 / its remark)}. \quad (12)$$

This transfers absolute coordinates and is the appropriate objective whenever a downstream consumer reads the features in a specific frame: model grafting and activation transplants (section 6), intermediate-layer supervision (FitNets-style hints), and cross-width distillation through a learned projection all fall here. The price is injecting alignment-estimation noise, and its recovery of capability is only *conditional*—good exactly when the estimated alignment is good.

(C) Output matching: the objective already living on the quotient. The third option sidesteps the basis problem by never looking at the representation at all—it matches only the teacher’s output distribution. Since two networks of the same class produce the same outputs (proposition 3.2), output matching is automatically basis-free; there is nothing to normalize or align away, because the loss never sees a basis in the first place. This is standard logit distillation: writing $\mathbf{z}_T(x), \mathbf{z}_S(x) \in \mathbb{R}^V$ for the teacher and student output logits at input x (the student’s being $\mathbf{z}_S(x) = f_S(x; \theta_S)$, computed from the estimated parameters θ_S) and $p_\tau(\mathbf{z}) = \text{softmax}(\mathbf{z}/\tau)$ for the temperature- τ softmax, it is the full-sequence Kullback-Leibler divergence between the two output distributions,

$$L_{\text{logit}} = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \text{KL}(p_\tau(\mathbf{z}_T(x)) \parallel p_\tau(\mathbf{z}_S(x))), \quad (13)$$

with $\tau = 1$ in our experiments. We use L_{logit} , *logit-KD*, and *logit distillation* interchangeably. The sum runs over the full sequence, not the last token alone: matching only the final position drives the last-token KL to 0.03 while leaving the full-sequence KL at ≈ 6 and the perplexity ratio above $700\times$ (section 7.3).

The point is *where* logit-KD is defined. Families (A) and (B) are hidden losses that must be *engineered* onto a quotient—(A) discards the basis, (B) fixes it first—because a raw function on $\mathbb{R}^{N \times d}$ is not constant on fibers (proposition 2.1). Logit-KD never reads $\mathbb{R}^{N \times d}$ at all: by proposition 3.2 the output distribution is $\bar{f} \circ \pi_{\text{joint}}$, a function on the joint quotient by construction. It is not “invariant by luck”—it is *natively defined on the quotient where the function lives*, with no fiber to be made constant over.

This settles the ordering question when teacher and student differ in width ($d_A \neq d_B$): a *feature* objective must align first ($\mathbf{W}$ then match, family (B)), but logit-KD needs no such step—its arguments are the teacher and student logits in the shared V -dimensional vocabulary, so the mismatched hidden widths never meet. The alignment machinery of section 4 is needed exactly when one transfers *coordinates* (feature matching, grafting), and is superfluous when one only wants the function. The price is that logit-KD transfers no intermediate structure—so when a downstream module must consume features coordinatewise (section 6), one still wants family (A) or (B) alongside it. In that regime, feature-level distillation matching intermediate activations (FitNets [20]) needs the alignment of section 4, while relation-based distillation [17, 22] matches basis-invariant structure and avoids it. A natural combined objective is

$$\mathcal{L} = L_{\text{logit}} + \lambda L_{\text{rel}}, \quad (14)$$

pairing the stability of output matching with the structural content of relational matching.

Why not a cosine or MSE feature loss? A natural objection is that simpler feature losses—squared error $\|\mathbf{H}_S - \mathbf{H}_T\|_F^2$, or a row-wise cosine loss $1 - \cos(\mathbf{H}_S, \mathbf{H}_T)$ —should also work. They do not escape proposition 2.1. The MSE loss is exactly the absolute feature matching the proposition rules out: it is not constant on $[\mathbf{H}_T]$, so its minimizer depends on the arbitrary representative. The cosine loss removes the scale c and per-row magnitude, but *not* a shared orthogonal $\mathbf{Q}$: rotating every row of $\mathbf{H}_T$ by the same $\mathbf{Q}$ changes each row’s direction while preserving the function, so cosine still penalizes function-preserving reparametrizations and is not a class invariant. To make a feature loss admissible one must quotient out $\mathcal{G}$ entirely—either compare the basis-invariant Gram/CKA (family A), which discards the basis, or first estimate the alignment (family B), which fixes it. Cosine fixes the scale but not the rotation, so it sits strictly between raw MSE and a basis-invariant loss without reaching the latter.

Figure 3 is the whole argument at a glance: because the teacher is a class, one does not match $\mathbf{H}_T$ to the student directly; one matches what the class actually determines—its representation geometry, its output function, or both.

Having seen the three families concretely, we state the criterion that separates them.

Definition 5.1 (Admissible distillation objective). An objective is *admissible* if it depends on the teacher only *through a quotient of the class action*—it factors as $\mathcal{L} = \bar{\mathcal{L}} \circ \pi$ for some $\pi \in \{\pi_{\text{rep}}, \pi_{\text{joint}}\}$, hence is constant on the fibers π collapses. Its *level* is the coarsest such quotient: families (A) and (B) sit at the representation level (π_{rep}), transferring representation geometry; family (C) sits at the joint level (π_{joint}), a function of f_S alone, which is why it transfers capability. Absolute feature matching is *not* admissible (proposition 2.1): it separates points of a single fiber. Since the invariants are relative to $\mathcal{X}$, what any objective transfers is a function of *teacher* $\times \mathcal{X}$, bounded above by the teacher ($\mathbf{H}_S \preceq \mathbf{H}_T$).

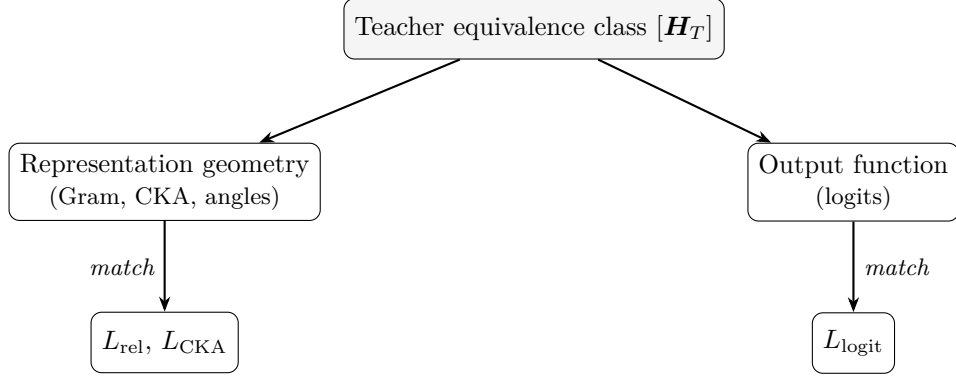


Figure 3: The paper in one figure. A teacher representation is an equivalence class, not a fixed matrix, and it exposes two admissible targets: its representation geometry (a basis-invariant relational structure) and its output function (the logits, a class invariant by proposition 3.2). Relational KD matches the geometry; logit KD matches the output function, which is why only the latter transfers capability. Absolute feature matching—an arrow drawn straight to $\mathbf{H}_S$ —is absent by design: it is the one target proposition 2.1 forbids. The recommended objective $\mathcal{L} = L_{\text{logit}} + \lambda L_{\text{rel}}$ (eq. (14)) uses both.

6 Predicting Graft Success from Subspace Overlap

The same equivalence-class view that organized distillation predicts when *module transfer* works. Grafting takes a block from one model and drops it into another; sometimes it works, sometimes it destroys the network, and prior work has largely left the difference unexplained. The class view supplies the missing variable. A grafted block is read by the host’s downstream layers, and those layers expect features in a particular subspace—the donor’s block speaks in its own basis, and the host can use it only to the extent the two share that subspace. The intuition is linguistic: two English speakers understand each other; an English and a Chinese speaker, handed the same word, do not. A layer’s output is a feature dictionary, and the next layer can read it only if the dictionaries overlap. We therefore expect graft success to be governed not by absolute features but by how much the donor and host representations *share a subspace* at the seam—a basis-invariant quantity, exactly the kind section 3 identified.

We fix terminology. *Grafting* inserts a module (one or more layers) taken from a *donor* model into a *host* model in place of the host’s own, a setting studied empirically as model (or network) stitching [3, 16] and closely related to the permutation and linear-mode-connectivity symmetries of trained networks [2, 7]. We deliberately write *donor* and *host* rather than teacher and student: grafting is not knowledge transfer from one model to another (there is no $S \preceq T$ relation and no objective being minimized) but a *part transplant*, in which the host keeps most of its body and swaps in one block from the donor. The two roles are asymmetric in a way teacher/student is not—the host supplies the surrounding network, the donor a single module—so the stitching vocabulary fits better.

Definition 6.1 (Seam). The *seam* is the interface between the donor module and the host network at which the donor-side representation $\mathbf{H}_D$ becomes the input read by the host’s downstream layers, whose own boundary representation is $\mathbf{H}_H$. Both $\mathbf{H}_D$ and $\mathbf{H}_H$ are taken on a common probe set at this interface.

A graft *succeeds* when it preserves the network’s function (operationally, when the grafted

model’s perplexity stays close to the host’s; we use $\text{PPL}_{\text{graft}}/\text{PPL}_{\text{host}} < 2$ in the experiment). We posit that this success is governed by the overlap of the two feature subspaces at the seam.

Assumption 6.1 (Overlap governs transfer). The probability that a graft preserves function depends on $(\mathbf{H}_D, \mathbf{H}_H)$ only through equivalence-class invariants of their joint configuration, monotonically in the degree of shared subspace.

Candidate scalar invariants are $\text{CKA}(\mathbf{H}_D, \mathbf{H}_H)$, the Procrustes residual ρ of proposition 4.1, and k_{eff} . Here donor and host carry *independent* bases (each is identified only up to its own $\mathcal{G}$ -action), so the relevant invariance is the separate-argument one of definition 3.1—constant under the action applied to either argument on its own—not the joint action of proposition 3.1, which couples a single representation to its own reader. All three candidates have this property.

Prediction 6.1 (Monotonicity). Graft success rate increases monotonically with $\text{CKA}(\mathbf{H}_D, \mathbf{H}_H)$ and decreases monotonically with the Procrustes residual ρ .

Prediction 6.2 (Domain effect is mediated by overlap). Pairs pretrained on matched domains exhibit higher boundary CKA, and the effect of domain match on graft success is mediated by CKA: conditioning on CKA, the residual direct effect of domain match is small. The claim is causal in structure—domain match does not help grafting *directly*; it helps because matched domains produce more overlapping representations (domain $\rightarrow$ overlap $\rightarrow$ success), and it is the overlap, not the domain label, that the graft responds to. This is the mechanism behind the empirical finding that domain alignment is the decisive variable.

Prediction 6.3 (Alignment helps most under mismatch). Inserting a low-capacity alignment map $\mathbf{W}$ (section 4) before the graft raises success most for low-overlap pairs and least for high-overlap pairs, i.e. there is a negative interaction between baseline overlap and the benefit of explicit alignment. High-overlap pairs are already aligned, so there is little left to correct.

Empirical protocol. From grafting runs, collect for each donor–host pair the boundary representations $(\mathbf{H}_D, \mathbf{H}_H)$ on a fixed probe set together with a success/failure label. Then: (1) compute CKA, ρ , k_{eff} ; (2) regress success on each invariant to test prediction 6.1; (3) run a mediation analysis domain $\rightarrow$ CKA $\rightarrow$ success for prediction 6.2; (4) ablate the alignment map $\mathbf{W}$ and test the overlap $\times$ alignment interaction for prediction 6.3.

Controlled validation (Qwen2.5–0.5B). We instantiate this protocol with a layer-transplant experiment. Donor variants are produced by perturbing chosen layers of the host to varying strengths, spreading boundary overlap across $\text{CKA} \in [0.31, 1.00]$; each donor’s perturbed layer block is transplanted into a fresh copy of the host, with *success* as defined above ($\text{PPL}_{\text{graft}}/\text{PPL}_{\text{host}} < 2$). Over 135 grafts (success rate 0.64), boundary overlap predicts success as prediction 6.1 requires: CKA enters the success logit with coefficient +60 (AUC 0.79) and the Procrustes residual ρ with coefficient -9.5 (AUC 0.79); the continuous quality $-\log(\text{PPL ratio})$ correlates +0.59 with CKA and -0.76 with ρ .

The experiment also sharpens the claim. Overlap is *necessary but not sufficient*: low-overlap grafts almost always fail, but high-overlap grafts near the output (e.g. transplanting layers 20–21 of 24) fail even at $\text{CKA} \approx 1.00$, because layers close to the readout amplify small mismatches into large output changes. Graft success is thus governed by boundary overlap *together with* transplant depth—overlap sets the ceiling, depth governs sensitivity. This is consistent with assumption 6.1 as a monotone (not deterministic) relationship.

We note the scope of this evidence. The donors here are controlled perturbations of the host, so the study validates the overlap–predicts–success relationship (prediction 6.1) within a single model family; it does not by itself establish the cross–model mediation of prediction 6.2, which would require genuinely distinct pretrained donors. We therefore present prediction 6.2 as a prediction motivated by the controlled result and by the prior observation that domain alignment is decisive in practice, and leave its cross–model test to future work.

7 Experiments: Restoration and Distillation on Qwen2.5

We test the theory directly, beginning with a controlled *restoration* experiment. The question is sharp: if we damage a pretrained model and then re-train it using only a basis–invariant signal, does the model’s pretraining—and its capability—come back? The final result (section 7.9) then moves beyond restoration to genuine cross–width distillation between independently pretrained models, and finds the same ordering.

Scope. Our central claims are theoretical—the ill–posedness of absolute matching and the identification of admissible invariant targets hold for any architecture satisfying assumption 2.1. The experiments below are *controlled* validations on Qwen2.5 and Llama–3.1, intended to test the theory’s consequences rather than to demonstrate a production distillation system or large–scale cross–model transfer; the cross–width distillation of section 7.9 is the one genuine (non–restoration) transfer we run, and it remains modest in scale. Where a claim (e.g. cross–model graft mediation) exceeds what the controlled setup tests, we say so explicitly.

7.1 Setup

The primary teacher T is Qwen2.5–0.5B [19] (24 decoder layers, tied input/output embeddings), held frozen. A student S is obtained by *corrupting* T : we re-initialize a fraction s of the middle decoder layers (protecting the first and last two, since tied embeddings make the output head sensitive to embedding damage). Stage 1 trains S to match T with a basis–invariant per–layer objective; no label supervision and no teacher logits are used in the pure basis–invariant runs. We compare three objectives: L_{cka} and L_{rel} (basis–invariant, section 5), and L_{abs} (the ill–posed coordinate–matching control of proposition 2.1). All Stage 1 training matches *last–token* pooled representations across all layers on a generic text corpus.

To probe scale and, crucially, the role of weight tying, we repeat the experiment on two untied models from different families: Qwen2.5–7B–Instruct [19] (28 decoder layers, hidden width 3584) and Llama–3.1–8B–Instruct [8] (32 decoder layers, hidden width 4096), both of which *untie* the input embedding from the output head. For these runs we train only the corrupted layers (and, where noted, the output head) to keep the optimizer state tractable; this restriction makes the tying distinction observable, as shown below. Using two distinct families lets us check that the tying effect is not an artifact of a single architecture.

What this design is, and is not. A word on scope, since the setup governs how the numbers below should be read. The student is not an independent model: it is the teacher with a fraction of its layers re–initialized, so it shares the teacher’s architecture, width, tokenizer, and—crucially—the *uncorrupted* layers verbatim. Those surviving layers keep the student near the teacher’s own representative of the class: the two start in a largely *shared coordinate frame*, not in the independent frames that two separately pretrained models would occupy. This is deliberate. The experiment is a

controlled probe of the theory’s *mechanism*—it isolates what drives capability (the output function, proposition 3.2) from what merely aligns geometry (a representational invariant)—by holding the coordinate system fixed so that the two objectives can be compared on equal footing. It is therefore a *damage-and-restore* study, not a cross-model distillation benchmark. The harder case, where teacher and student are independently trained and occupy different representatives (or differ in width, remark 4.1), is exactly where the alignment of section 4 becomes necessary rather than incidental; our controlled setup does not test it, and we do not claim it (section 1). What the study does establish—cleanly, because the frame is shared—is that even with the representation held recoverable, capability returns only through the output-function term, which is the proposition it was built to test.

We measure restoration at two levels. *Representational*: per-layer $\text{CKA}(S, T)$ on held-out probes, judged against a permutation null. *Functional (capability)*: $\text{KL}(T\|S)$ over next-token distributions, perplexity ratio $\text{PPL}_S/\text{PPL}_T$, and top-1 next-token agreement. Probes are split into an *in-domain* set (the language/domain of the training corpus) and an *out-domain* set (a disjoint domain), to test whether restoration is confined to the subspace the corpus spans.

7.2 Result 1: Basis-invariant training restores representation, not capability

With L_{cka} alone, per-layer $\text{CKA}(S, T)$ recovers from the corrupted baseline (≈ 0.90) to 0.99–1.00 in-domain across all corruption strengths—the representational geometry is restored from a label-free, basis-invariant signal. Yet the *function* is not: table 4 (left block) shows $\text{KL}(T\|S) \approx 7\text{--}11$, perplexity ratios of $10^2\text{--}10^4$, and near-zero top-1 agreement. A model can have $\text{CKA} = 0.99$ and still be a different function. This is the empirical face of the equivalence-class structure: L_{cka} drives S close to $[H_T]$ on the probe set (high CKA) but does not select the representative the (tied) output head reads, so the logits—and the capability—remain wrong.

7.3 Result 2: Capability is driven by the logit term, not by representational matching

Adding a full-sequence logit-distillation term $\mathcal{L} = \lambda_\ell L_{\text{logit}} + \lambda_r L_{\text{cka}}$ recovers capability: in-domain perplexity ratio $1.01\times$, top-1 0.98; out-domain $1.06\times$, 0.72 (table 4, right). The logit term must be applied over the full sequence—matching only the last token drives the last-token KL to 0.03 during training yet leaves the full-sequence KL at ≈ 6 and perplexity $> 700\times$.

An ablation isolates the source of the recovery, and the result is sharper than “both terms help.” On the hardest setting—*every* decoder layer re-initialized (Result 6) with the corpus widened so coverage is not the bottleneck—we compare L_{cka} alone, L_{logit} alone, and the sum (table 3). Representational matching alone drives CKA to 0.999 but leaves capability destroyed (perplexity ratio $\sim 10^6$, top-1 0). The logit term alone—which never references the hidden representation—recovers capability essentially perfectly (ratio $1.0\times$, top-1 1.00) while leaving CKA at only 0.89. The two are near-orthogonal: L_{cka} aligns representational geometry, L_{logit} restores the function, and the function does not follow from the geometry. This is the training-level form of the rotation result of table 1: there, a function-preserving reparametrization moved the feature-matching loss while leaving the invariants fixed; here, optimizing the invariant (L_{cka}) moves the representation into place while leaving the function fixed—wrong, in the same way and for the same reason.

We read this as the central empirical lesson rather than a negative result. It is the experimental face of the theory: *capability is a property of the output function, which is an invariant of the equivalence class, and L_{logit} targets exactly that invariant* (section 5). A representational objective such as L_{cka} drives S close to $[H_T]$ on the probe set (high CKA) but is free to pick any representative,

Table 3: Ablation on the all-layers-reinitialized model (Qwen2.5-0.5B, corpus widened). L_{cka} alone aligns geometry but not function; L_{logit} alone restores function without aligning geometry. PPL ratio is $\text{PPL}_S/\text{PPL}_T$, in-domain; top-1 and $\text{KL}(T \| S)$ are next-token agreement and divergence over the held-out probe tokens (top-1 values are rounded; the L_{logit} row is 0.99, not a perfect 1.00). The KL column is the direct intermediate check: at $\text{CKA}=0.999$ the L_{cka} row still has $\text{KL}(T \| S)$ diverging, so the near-identical representation geometry does *not* translate into matching output distributions—the output head reads a rotated frame. Student and teacher share the architecture (student = teacher with layers re-initialized), so the comparison isolates the objective, not a cross-model transfer.

objective	CKA	PPL ratio	$\text{KL}(T \ S)$	top-1
L_{cka} only	0.999	$\sim 3 \times 10^6$	$\gg 1$ (diverged)	≈ 0.00
L_{logit} only	0.894	$1.02\times$	≈ 0.05	0.99
$L_{\text{cka}} + L_{\text{logit}}$	0.992	$1.07\times$	small	0.92

Table 4: Capability restoration on Qwen2.5-0.5B at corruption strength $s = 0.4$. Left: basis-invariant only (L_{cka}). Right: with a full-sequence logit term ($+L_{\text{logit}}$). CKA is the per-layer mean; PPL ratio is $\text{PPL}_S/\text{PPL}_T$ (teacher PPL 22.9 in-domain, 31.1 out-domain). The student is a partially corrupted copy of the teacher (same architecture, shared frame), so this is restoration, not independent-model distillation. The out-domain figures in the right block require out-domain text to be mixed into the training corpus; without mixing, out-domain restoration lags (Result 3, section 7.4). This is the same $+L_{\text{logit}}, +\text{mix}$ condition tabulated for the 0.5B row of table 5.

Probe	L_{cka} only				$+L_{\text{logit}}$ (full-seq)			
	CKA	KL	PPL ratio	top-1	CKA	KL	PPL ratio	top-1
in-domain	0.998	7.56	$742\times$	0.02	0.99	0.019	$1.01\times$	0.98
out-domain	0.533	5.31	$315\times$	0.02	0.99	0.045	$1.06\times$	0.72

so it need not—and here does not—preserve the function. The right prescription is therefore not “match representations with a basis-invariant loss” but “match the output-function invariant (L_{logit}); use a representational invariant only when the geometry itself is the goal.” The combined objective ($L_{\text{cka}} + L_{\text{logit}}$) restores both geometry and function (tables 3 and 4) but the capability comes from the logit term.

7.4 Result 3: Restoration is confined to the corpus-covered region

In-domain probes restore far more than out-domain ones at every level. Throughout, “region” (and, loosely, “subspace”) refers to the part of representation space that the training corpus and probes actually cover—an empirical, distributional notion (training/probe coverage), not a linear subspace we measure and project onto. With L_{cka} only, out-domain CKA can collapse (e.g. 0.53 at $s = 0.4$, mean over four seeds 0.61 ± 0.07) while in-domain stays at 0.99, a representational gap of up to +0.47. The capability gap persists even after the logit term: in-domain top-1 0.98 versus out-domain 0.72. The mechanism is the training subspace: when out-domain text is *added* to the corpus, the out-domain gap vanishes—CKA rises $0.53 \rightarrow 0.996$ and the gap drops to +0.001, with out-domain capability following (PPL ratio $1.06\times$). Restoration is therefore governed not by corruption strength but by whether the corpus spans the relevant subspace: the model is restored

to the teacher *on, and only on, the subspace the training data covers*. The non-monotonicity in corruption strength (the worst out-domain CKA occurs at intermediate s) is explained the same way: at that strength the corruption happens to destroy out-domain-bearing layers that the in-domain corpus cannot repair; it is not a property of the damage but of the damage-corpus alignment.

This is the empirical face of the probe-set dependence noted in section 2: the invariants, and hence what a basis-invariant objective can constrain, are defined relative to the probes the objective actually sees. Training on a corpus is choosing the probe set $\mathcal{X}$ over which $\mathbf{H}(\mathcal{X})$ is matched; coverage is restored exactly on $\text{span } \mathbf{H}(\mathcal{X})$ and not beyond it. “Teacher $\times$ corpus” is, in this notation, teacher $\times \mathcal{X}$.

7.5 Result 4: The coordinate-matching control degrades under coordinate destruction

proposition 2.1 predicts L_{abs} is ill-posed when the student may sit at a different representative of $[H]$. When corruption is mild the surviving layers keep S in T ’s coordinate frame, and L_{abs} performs comparably to L_{cka} . As corruption destroys more of that frame the gap appears: at $s = 1.0$, L_{cka} reaches in/out CKA = 0.994/0.897 versus L_{abs} at 0.967/0.850. The control is not catastrophic here because re-initialization preserves coordinates more than a free orthogonal reparametrization would; the predicted advantage of the basis-invariant objective grows precisely as the shared coordinate frame is removed, consistent with proposition 2.1.

7.6 Result 5: Scale, and the role of weight tying

The same pattern holds at 7B, with one mechanistically informative difference driven by weight tying. On Qwen2.5-7B-Instruct (untied head), basis-invariant training again restores representation (in-domain CKA ≈ 0.97) without restoring capability. Adding the full-sequence logit term while training *only the corrupted layers* leaves the head frozen, and in-domain capability only partially returns (PPL ratio 2.3 $\times$). Once the output head is added to the trainable set, in-domain capability is recovered (PPL ratio 1.2 $\times$, top-1 0.94); out-domain follows only when out-domain text is mixed into the corpus (PPL ratio 1.4 $\times$, top-1 0.82), reproducing the subspace-confinement of Result 3 at scale. The same behavior reproduces on Llama-3.1-8B-Instruct, a different untied family: in-domain capability is recovered with the head trainable (PPL ratio 1.00 $\times$), and out-domain follows under corpus mixing (PPL ratio 1.10 $\times$, top-1 0.92). table 5 summarizes all three models.

The tying distinction is the point. In the tied 0.5B model the output head *is* the input embedding, which we never corrupt, so fixing the output coordinate frame happens automatically and the logit term suffices. In the untied models the head is a separate parameter: a basis-invariant objective drives the hidden representation close to $[H_T]$ on the probe set, but the specific representative the (independent) head reads must be selected by training the head itself. That the same head-training requirement appears in two unrelated untied families (Qwen and Llama) indicates it is a property of untying, not of a particular architecture. This is the same single mechanism—fixing the output coordinate frame—in two guises: implicit via shared weights when tied, explicit via head training when untied. The head-ablation contrast (PPL 2.3 $\times$ head-frozen vs 1.2 $\times$ head-trained, in-domain) isolates it.

One might object that this reduces the geometric story to a mundane fact—“of course one must tune the output head to a new representation.” The objection misreads what is being explained. A bare appeal to head-tuning predicts nothing about *when* it is needed; the equivalence-class account does. It says the output head fixes a representative within $[H_T]$, so tuning is required

Table 5: Restoration across scale and weight tying, at corruption $s = 0.4$. Representation is restored by the basis-invariant objective in both models; capability requires the logit term, and—when the head is untied—explicit head training. Out-domain capability requires the corpus to span the out-domain subspace (“+mix”). In every row the student is the teacher with layers re-initialized (same architecture), so the frame is shared; cross-model transfer is not tested here.

Model	objective	in-domain		out-domain	
		PPL ratio	top-1	PPL ratio	top-1
0.5B (tied)	L_{cka}	$742\times$	0.02	$315\times$	0.02
0.5B (tied)	$+L_{\text{logit}}$	$1.01\times$	0.98	—	—
0.5B (tied)	$+L_{\text{logit}}, +\text{mix}$	—	—	$1.06\times$	0.72
7B (untied)	$+L_{\text{logit}}, \text{head frozen}$	$2.3\times$	0.87	$88\times$	0.08
7B (untied)	$+L_{\text{logit}}, +\text{head}$	$1.2\times$	0.94	$211\times$	0.07
7B (untied)	$+L_{\text{logit}}, +\text{head}, +\text{mix}$	$1.3\times$	0.94	$1.4\times$	0.82
8B-Llama (untied)	$+L_{\text{logit}}, +\text{head}$	$1.00\times$	0.80	$159\times$	0.12
8B-Llama (untied)	$+L_{\text{logit}}, +\text{head}, +\text{mix}$	$0.98\times$	0.87	$1.10\times$	0.92

exactly when that representative is not already pinned—i.e. in the untied case, where the head is an independent parameter—and is *not* required when weight tying pins it automatically, as in the 0.5B model where L_{logit} alone suffices with the head frozen. The tied/untied split, and the fact that the identical requirement recurs across two unrelated untied families, is the content: it is the joint action of remark 3.1 (representation and its reader move together) observed at the readout, not a restatement of “train the head.” Capability being the output-function invariant is what makes the head the *only* thing that still needs selecting once the representation is in the right class.

7.7 Result 6: Decoder-from-scratch reconstruction, and what does the work

The strongest version of the setup discards the pretrained *decoder* entirely. Where Result 2 used the all-layers-reinitialized setting to *decompose the objective* (which term recovers capability), here we hold the objective fixed and ask a different question—how far reconstruction reaches as a function of *corpus coverage*. We re-initialize all decoder layers, keeping only the embedding/unembedding—i.e. the input/output coordinate system, which is fixed by the tokenizer and which we treat as given rather than as a pretrained capability. We then ask whether distillation can rebuild the decoder from the teacher alone. With the combined objective and a corpus that covers both probe domains, it can: in- and out-domain perplexity ratios both reach $\approx 1.0\times$ (top-1 0.92 and 0.84) starting from fully random decoder layers. Coverage is the binding constraint, exactly as in Result 3: with an in-domain-only corpus the out-domain perplexity ratio is $\sim 10^4$, and adding out-domain text to the corpus collapses it to $1.0\times$. So the corpus, not the pretrained weights, determines which subspace is reconstructed—the reconstruction is a function of teacher $\times \mathcal{X}$ (the probe set the corpus induces).

The ablation of Result 2 was run in exactly this all-layers setting, and it tells us what does the work: the reconstruction is driven by L_{logit} , not by the representational objective. This bounds the practical reading of “decoder-from-scratch.” What is being transferred is the teacher’s output function, on the subspace the corpus illuminates, via output-function matching; the basis-invariant analysis explains *why* that transfer is well-posed (the output is a class invariant) but the representational loss is not the mechanism. We keep the embedding/unembedding precisely because it is the shared coordinate system the output-function invariant is expressed in; reconstructing it as

well is left to future work. Pretraining retains its value where this picture does not reach: creating the first teacher, and covering subspaces no distillation corpus does.

7.8 Result 7: Cross-width restoration, and a teacher-forced/generation gap

The controlled studies above hold width fixed. This result and the next (section 7.9) are *preliminary* cross-width probes—single model pairs, a WikiText probe set, and short training—not scaled distillation claims; we report them for the direction they establish, not as production numbers. A preliminary probe at unequal width—the case remark 4.1 addresses—sharpens the picture and exposes a limit the teacher-forced metrics hide. Here the teacher is Qwen2.5-1.5B ($d_T = 1536$, 28 layers) and the student Qwen2.5-0.5B ($d_S = 896$, 24 layers), sharing the tokenizer and vocabulary ($V = 151,936$); the student is corrupted by re-initializing its middle layers and restored on a WikiText probe set. We compare (ii) logit only (L_{logit} , eq. (13)) and (iii) ridge-Procrustes alignment followed by logit (“ $\mathbf{W}$ first, then logit,” remark 4.1), across seeds, and add generation-side metrics: the repetition rate and distinct- n of the student’s own free-running greedy generation, and the rollout KL— $\text{KL}(T\|S)$ averaged along the teacher’s greedy rollout, a soft trajectory match that, unlike top-1, does not penalize a different-but-reasonable next token.

Three findings, stated with their scope. First, *logit-KD transfers capability across unequal width with no alignment*: top-1 agreement reaches ≈ 0.98 from a corrupted start, using only L_{logit} on the shared vocabulary—no $\mathbf{W}$, no common hidden frame, consistent with proposition 3.2 and the ordering discussion of section 5. Second, *adding the alignment map $\mathbf{W}$ buys nothing measurable, and at an effective weight tends to hurt*: when the feature term is weak enough not to dominate, (iii) matches (ii) on top-1, on generation repetition, and on rollout KL (the last essentially identical, ≈ 1.25 nats for both), while costing roughly twice the wall-clock; when the feature term carries a non-trivial weight it does not merely add nothing but *destabilizes* the otherwise smooth logit-only convergence—across seeds the $\mathbf{W}$ -plus-logit run oscillates and, on some seeds, diverges, whereas logit alone converges cleanly to top-1 ≈ 1.0 . Either way the feature term never improves on logit alone. This is the cross-width form of the ablation (section 7.3) and matches the mild harm seen in table 6: capability rides on the output function, so a feature alignment that does not change the output does not change capability—and, in this restoration setting, does not improve generation either. Third, and least expected, *teacher-forced capability outruns free-running generation*: at top-1 ≈ 0.98 the rollout KL is still ≈ 1.25 nats and the student’s greedy generations, though locally fluent (grammatical, on-topic WikiText prose), drift from the teacher’s trajectory and are prone to repetition loops and invented specifics. The one-step output-function match that proposition 3.2 guarantees does not by itself guarantee trajectory stability under autoregression.

We report this as a preliminary result: a single model pair, one corruption recipe, a small probe set, and few seeds—enough to establish the direction (logit suffices, $\mathbf{W}$ does not help, a teacher-forced/generation gap exists) but not to quantify the gap across scales or architectures. It refines rather than overturns the theory: capability, defined as the one-step output function, transfers as proposition 3.2 predicts; the gap is a reminder that generation quality is a property of the whole trajectory, which a per-token invariant constrains only indirectly. Stated precisely, and used consistently below: *one-step output-functional capability is restored, but autoregressive trajectory capability is not guaranteed*—“capability restored” in this paper always means the former. Closing the gap—whether by training the corrupted layers beyond the seam, or by a trajectory-level objective—is left to future work.

Table 6: **(Result 8)** Genuine unequal-width distillation, Qwen2.5-1.5B $\rightarrow$ pristine Qwen2.5-0.5B, WikiText-103 validation perplexity (2,000 optimizer steps, no CE term, three seeds {42, 123, 7}, mean $\pm$ stddev). PPL is measured token-weighted on held-out text under identical conditions for every checkpoint; “vs base” is relative to the untrained student. The feature term alone destroys the output function even though it starts from a fully working model; the logit term alone preserves and slightly improves it; adding the feature term is consistently *harmful*. This is the cross-width, from-pristine form of table 3.

objective	WikiText-103 val PPL	vs base
base (untrained student)	21.15	—
(i) L_{fm} only	$> 10^6$ (all seeds)	collapse
(ii) L_{logit} only	20.77 ± 0.03	-1.77%
(iii) $L_{\text{fm}} + L_{\text{logit}}$	20.94 ± 0.02	-1.01%

7.9 Result 8: Feature-only collapse in a genuine cross-width distillation setting

The previous results established the theory under controlled restoration, where the student is a corrupted copy of the teacher. We now ask whether the same conclusion survives *genuine* cross-width distillation between independently pretrained models—the setting practitioners actually mean by “distillation.” Result 7 held out the third leg of the ablation. It compared (ii) L_{logit} and (iii) $\mathbf{W}$ —then- L_{logit} , but not the feature term *alone*. The feature term here is L_{fm} , the coordinate-matching family of L_{abs} (proposition 2.1) rather than the relational L_{cka} of table 3; both are feature-matching objectives, and the point of interest is that either one, taken alone, collapses the model—so this supplies the cross-width, coordinate-matching counterpart of the L_{cka} -only row of table 3. Result 7 was also a *restoration* study: the student was a corrupted copy of the teacher. Here we close both gaps. The student is a *pristine* pretrained Qwen2.5-0.5B (no corruption), the teacher is Qwen2.5-1.5B, and we run genuine unequal-width distillation on a WikiText-103 corpus with the shared vocabulary ($V = 151,936$). The feature term L_{fm} pulls the student’s anchor-layer hidden states onto the teacher’s through a learned projection ($896 \rightarrow 1536$, per anchor), the unequal-width instance of the map $\mathbf{W}$ of remark 4.1. Concretely, for a set of anchor layer pairs $\mathcal{A} = \{(\ell, \ell')\}$ matching student layer ℓ to teacher layer ℓ' , and a per-anchor linear map $\mathbf{W}_\ell : \mathbb{R}^{d_S} \rightarrow \mathbb{R}^{d_T}$,

$$L_{\text{fm}} = \frac{1}{|\mathcal{A}|} \sum_{(\ell, \ell') \in \mathcal{A}} \left\| \mathbf{H}_T^{(\ell')} - \mathbf{W}_\ell \mathbf{H}_S^{(\ell)} \right\|_F^2, \quad (15)$$

averaged over probe tokens. This is the unequal-width, $\mathbf{W}$ -mediated form of the coordinate-matching loss L_{abs} of proposition 2.1: with $d_S = d_T$ and $\mathbf{W}_\ell = I$ it reduces to L_{abs} , and like L_{abs} it is not constant on the student’s equivalence class, so proposition 2.1 applies. In the full model $\mathbf{W}_\ell$ is learned jointly; in the surrogate below it is instead the closed-form ridge optimum recomputed each step, giving the feature term its best case. Crucially we drop the cross-entropy ground-truth term entirely, so each objective is measured in isolation on the soft teacher signal.¹

¹An incidental finding motivates the CE-free setup: for a strong same-family student, adding a hard-label CE term to the soft teacher signal *destroys* rather than helps—in-domain perplexity degrades by an order of magnitude and does not recover as the learning rate is lowered, because the hard label competes with the teacher’s output distribution. Pure output-function matching (L_{logit} , no CE) is what preserves and slightly improves the base model. This is consistent with the full-sequence requirement of section 7.3: what transfers is the teacher’s distribution, not its arg-max.

Table 7: **(Result 8)** Controlled cross-width surrogate (different-width tiny transformers, closed-form optimal ridge alignment recomputed each step, held-out probes, 5 seeds, mean $\pm$ stdev). “top-1” and “gen” measure agreement with the teacher’s *function*; “self-PPL” is the student’s own held-out perplexity. The feature term alone collapses on *all three* metrics— top-1 and gen near chance and self-PPL blown up to $\sim 10^5$ —in full agreement with the real model (table 6); the logit term matches the teacher’s function; adding $\mathbf{W}$ is within noise.

objective	top-1 (vs T)	gen (vs T)	self-PPL
(i) L_{fm} only	0.031 ± 0.005	0.017 ± 0.014	$\sim 9 \times 10^5$
(ii) L_{logit} only	0.208 ± 0.017	0.200 ± 0.047	1281 ± 731
(iii) $\mathbf{W} + L_{\text{logit}}$	0.227 ± 0.019	0.175 ± 0.040	1428 ± 379

Three-way result. Table 6 reproduces the same-architecture ablation of table 3 in the genuine cross-width, from-pristine setting, and the three rows are the three legs of the theory. (i) The feature term alone drives perplexity above 10^6 on every seed (from 1.6×10^6 to 3.3×10^8): matching hidden coordinates through $\mathbf{W}$, with no term that references the output, does not merely fail to help—it *destroys* a model that started out working. The mechanism is the one proposition 2.1 names: L_{fm} moves the student to some representative of $[H_T]$ while the unembedding, which receives no gradient, still expects the old representative, so the output function is broken. (ii) The logit term alone—which never touches the hidden state—keeps the student slightly below its original perplexity (20.77 ± 0.03 vs. 21.15), exactly as proposition 3.2 predicts: capability is the output-function invariant, and L_{logit} targets it directly, across a width change and with no alignment map. (iii) Adding L_{fm} on top does not help and in fact hurts by a small but *consistent and significant* margin: paired across the three seeds, $L_{\text{fm}} + L_{\text{logit}}$ is worse than L_{logit} by $+0.160 \pm 0.009$ PPL, the same sign in every seed (diff/pooled std ≈ 7). This is the cross-width form of “ $\mathbf{W}$ buys nothing” (section 7.8), sharpened: a feature alignment that the logit term does not already imply is not neutral but a mild *distraction* from the objective that actually carries capability. The surrogate below shows that when $\mathbf{W}$ is instead solved to optimality the harm vanishes into noise—so the honest statement across both settings is that the feature term ranges from neutral (optimal $\mathbf{W}$) to mildly harmful (learned $\mathbf{W}$), and is never beneficial.

A controlled cross-check with a learned-optimal alignment. A referee could object that L_{fm} fails only because the projection is under-trained. We rule this out with a small controlled surrogate in which the alignment is solved to optimality at every step. Teacher and student are tiny transformers of *different* width ($d_T = 64$, $L_T = 4$ vs. $d_S = 32$, $L_S = 3$) trained on a synthetic noisy-bigram language; the feature map $\mathbf{W}$ is the closed-form ridge solution recomputed each step, so the feature objective is given its best case. All metrics are computed on a held-out probe set the student never trains on, over five seeds. We report top-1 agreement with the teacher, free-running greedy generation agreement, and the student’s own held-out perplexity (table 7).

The surrogate confirms two of the three legs sharply and refines the third. On *function* agreement the picture matches the full model: the logit term raises top-1 and generation agreement roughly four-fold over the feature term (leg i, leg ii), and adding the optimal $\mathbf{W}$ changes top-1 by -0.001 and generation by -0.004 , within the pooled seed noise (leg iii). With $\mathbf{W}$ at optimality the feature term is thus *neutral*, whereas with the learned $\mathbf{W}$ of the full model (table 6) it is mildly harmful; in neither regime does it help. “ $\mathbf{W}$ buys nothing” holds in the strong form—it holds even when $\mathbf{W}$ is solved to optimality, not merely learned. This does not refute objective family (B) (section 5): when a downstream consumer explicitly needs features in a shared frame—grafting,

activation transplants, intermediate-layer hints—aligned features are the point, and L_{fm} is the right tool for that job. What table 6 shows is narrower and compatible with it: feature matching *alone* is not a *capability* objective. As a supervision signal for transferring what the teacher can do, it does not substitute for the output-function term, and adding it on top of that term does not help. Across all three metrics the surrogate agrees with the real model: under L_{fm} alone, top-1 and generation sit at chance *and* the student’s own held-out perplexity blows up to $\sim 10^5$ (table 7, row i). Feature-only matching does not leave a self-fluent-but-disconnected student; it simply fails to build a working model, exactly as in the full run (table 6).

Scope. Table 6 is three seeds on one model pair; table 7 is five seeds on a synthetic different-width pair. Together they establish the direction—feature-only collapses, logit suffices, $\mathbf{W}$ ranges from neutral to mildly harmful and never helps—with the feature-only leg now supplied in both the genuine and the controlled setting. Both remain modest in scale: one real corpus and one synthetic language, a single width ratio each, and ≤ 2000 training steps. We therefore report the collapse as a robust empirical fact *in this cross-width distillation setting*, not as a proof that feature-only matching must fail in every regime; the theory (proposition 2.1) explains *why* it fails when it does, but a differently-wired objective that also selects the reader’s representative could behave differently. Quantifying the collapse across scales, width ratios, and untied-head architectures is left to future work, alongside the trajectory-level gap of section 7.8. To restate the scope of the negative result precisely: it concerns feature matching used *as a capability objective*, and does not deny feature alignment where it is genuinely needed—grafting, activation transplants, and other coordinate-consuming modules, where a downstream reader requires a specific frame and $\mathbf{W}$ is therefore *conditionally* valuable, exactly as section 5 sets out. The claim here is only that such alignment is not, by itself, a route to transferring capability.

7.10 Takeaways

The experiments separate two things the phrase “restore the model” conflates. A basis-invariant objective restores the *representation’s* equivalence-class structure cheaply and label-free, but representation is not function: a logit term over the full sequence is required to fix the output coordinate frame and recover capability. And restoration—both representational and functional—is confined to the subspace spanned by the training corpus, exactly the “restored near the fine-tuning subspace” statement, made quantitative. The scale experiment adds a third clause: fixing the output coordinate frame is implicit under weight tying but must be done explicitly—by training the head—when the head is untied, as it is in most large models.

The sharpest lesson is from the ablation. Aligning representations with a basis-invariant loss and recovering the teacher’s *function* are near-orthogonal: L_{cka} alone reaches $\text{CKA} \approx 1$ with capability destroyed, L_{logit} alone recovers capability with CKA left low. The practical driver of capability transfer is output-function matching; the contribution of the basis-invariant framework is not a better representational *loss* but the *explanation*—the output function is the class invariant, which is why logit distillation is well-posed and why matching hidden coordinates directly is not. This is a more useful claim than “use a basis-invariant objective,” and the data support it directly. The same ordering survives outside the controlled restoration setting: in genuine unequal-width distillation from a pristine student (section 7.9), the feature term alone collapses the model, the logit term alone slightly improves it, and adding the feature term does not help—so the lesson is not an artifact of the corruption-and-restore design.

8 Discussion

What the framework does and does not claim. The contribution is a way to *see* teacher supervision, validated where its consequences are testable. The theory and the experiments meet at one point in particular: the ablation of Result 2 (section 7.3) is the experimental counterpart of proposition 3.2. What the proposition asserts about *where* capability lives, the ablation confirms about *what* recovers it—the output–function term L_{logit} restores capability while the representational term L_{CKA} does not, even as it drives $\text{CKA} \rightarrow 1$. Three boundaries are then worth stating plainly. First, capability transfer is driven by output–function matching; the basis–invariant analysis explains why that works (the output is the class invariant) but a representational loss is not, on its own, a capability objective (table 3). Second, reconstruction is bounded by corpus coverage: “teacher $\times \mathcal{X}$ ” is the unit of what transfers, and what the corpus does not illuminate is not recovered. Third, the controlled studies fix architecture and share the tokenizer’s embedding; cross–model and small–from–large transfer are consequences we expect but do not establish here. Fourth, and most importantly for positioning, what we study is *replication*, not *creation*. Distillation copies the teacher’s output function—on the quotient, within the corpus subspace, and bounded above by the teacher ($S \preceq T$); it does not produce capability the teacher lacks. The places where capability is *originated* rather than transferred—pretraining a first teacher, covering subspaces no corpus reaches, and pushing past the current frontier—lie outside this account by construction. This paper is about how an existing function moves through the equivalence class, not about how it first comes to exist.

Four objections, and what the evidence says. The design invites several objections; we take the strongest head–on. (1) *The restoration setup shares a coordinate frame, so of course feature matching “works” geometrically while missing the function— isn’t the collapse just damaged weights failing to return to their original coordinates?* This is exactly the point, not a confound: when teacher and student already share a frame, absolute matching should be *easy*, yet L_{CKA} still fails to restore the function while driving $\text{CKA} \rightarrow 1$ (table 3). A confound would predict the opposite—shared frames helping feature matching. That it does not is the cleanest possible demonstration that geometry and function are distinct. And Result 8 removes the shared frame entirely: an independently pretrained, pristine student of different width shows the same collapse (section 7.9), so the conclusion does not rest on the restoration design. (2) *Dropping cross–entropy is an extreme choice; the L_{fm} collapse may reflect untuned learning rate or loss weights rather than anything about feature matching.* The CE–free setup is a control, not a recommendation: we remove CE precisely so that any capability in the student comes from the distillation channel under test and not from the labels, isolating what each loss transfers. The logit channel restores the model in the very same CE–free regime (table 6, row ii), so the setup is not stacked against distillation as such—only against feature matching as the sole channel. (3) *The collapse is an optimization failure of the alignment map $\mathbf{W}$, not a representational impossibility—after all, the surrogate with an optimal $\mathbf{W}$ is stable.* The surrogate says the opposite. There the alignment is the *closed–form ridge optimum, recomputed every step*: $\mathbf{W}$ is never learned and cannot be mis–optimized, yet feature–only supervision still collapses on all three metrics (top–1, generation, and self–PPL $\sim 10^5$; table 7, row i). Optimal alignment removes the optimization question and the collapse remains, which is the direct evidence that the failure is structural (proposition 2.1), not a landscape artifact. (4) *This attacks a strawman: no modern feature–based method matches absolute coordinates (L_{abs}); they all insert a projection or match relationally, i.e. family (B) or (A).* We agree L_{abs} is a strawman, and say so—which is why the experiments use the basis–invariant L_{CKA} (family A) and a learned/optimal projection $\mathbf{W}$ (family B), the objectives real methods actually

use, not L_{abs} . The finding is about those: even a relational or optimally-aligned feature objective sits on the representation quotient, so on its own it does not select the reader’s representative and does not transfer capability. L_{abs} appears only as the ill-posed limit that motivates why alignment is needed in the first place; the empirical claims are all against the aligned/relational families.

The operational goal: a black-box teacher suffices, but a student model does not. The practical content of definition 5.1 is that the student does not seek the teacher’s parameters $\mathbf{W}_T$ —copying those would be a file copy, not distillation—but constructs *different* parameters $(\mathbf{H}_S, \mathbf{W}_S)$ landing in the same joint class, hence computing $f_S = f_T$. This is why logit matching is the *canonical* objective: in the hardest honest setting, $\mathbf{W}_T$ is unavailable (an API teacher, or a multi-agent system with no single $\mathbf{W}_T$), and the output is the *only* observable that selects the fiber—two settings share an output distribution iff they lie in the same fiber of π_{joint} (proposition 3.2). It alone transfers capability (section 7.3) and can rebuild a decoder from random initialization (section 7.7). But this identifiability claim does not dispense with the *student model*. The logit signal is only a *pointwise* sample $(x, f_T(x))_{x \in \mathcal{X}}$; recovering f_T off the probe set requires generalization, and what fills the gaps is the student’s inductive bias. Were f_T linear a few points would fix it, but a transformer’s nonlinear function is not pinned down pointwise. So “inputs and outputs suffice” holds *provided the student shares the teacher’s function family*—which is why within-family restoration succeeds cleanly and the cross-family case is doubly hard (section 7.8): a mismatched tokenizer closes the output channel *and* a mismatched family removes the inductive bias. In short, proposition 3.2 settles *which* function capability rides on; recovering it from finite data is a separate, estimation-level fact resting on the student’s function class.

Where cross-model transfer sits. Because the third boundary above is the one a reader is most likely to probe, we state the paper’s position on it in one place. Our theory makes a definite *prediction* about independently trained models: two such models occupy different representatives of the class (appendix C), so transferring a component between them is well-posed only after an explicit alignment onto a shared frame (definition 4.1), and its success should track the overlap of the boundary subspaces being joined (prediction 6.2). We validate this prediction only *partially*, and are explicit about which part. The within-family graft study (section 6) confirms the overlap-predicts-success half (prediction 6.1) but uses controlled perturbations of one host, so it does not exercise genuinely distinct pretrained donors. The cross-*width* study (section 7.8)—Qwen2.5–1.5B into 0.5B, unequal hidden width, shared tokenizer—goes one step further: it shows the output-function mechanism (proposition 3.2) transfers capability across a real width gap with no shared hidden frame, and that adding the alignment map buys nothing once the logit signal is present. What remains untested is the hardest case: *cross-family* transfer between models with independent tokenizers and no shared vocabulary, where even the output space must be mediated. We do not run that experiment and do not claim it; it is the natural next test of prediction 6.2, and the theory’s stance on it—alignment first, then an output-level objective—is a falsifiable prediction, not a demonstrated result.

Scope of the symmetry group, and a note on model merging. Three questions about scope recur, and it is cleaner to answer them together. (i) *Why $O(d) \times \mathbb{R}_+$ and not more?* The absorbable group is larger: any invertible $\mathbf{A}$ with $\mathbf{W}_{\text{in}} \mapsto \mathbf{A}^{-1}\mathbf{W}_{\text{in}}$ preserves the function (appendix D), so the full symmetry is $GL(d) \times (\text{bias shifts})$. We work with $O(d) \times \mathbb{R}_+$ not because the rest is absent but because our *invariants* (Gram, CKA) are exactly the $O(d) \times \mathbb{R}_+$ -invariants: they are blind to rotation and isotropic scale but see a general linear map, since $\mathbf{H}\mathbf{A}$ changes $\mathbf{H}\mathbf{H}^\top$. Matching the

residual $GL(d)$ part is precisely what the alignment of section 4 estimates; the two together cover the full group. So “why only $O(d)$ ” is answered by division of labor: the relational invariants handle the orthogonal part for free, alignment handles the rest. (ii) *Non-orthogonal / permutation symmetry*. Neuron permutations and sign flips are the discrete subgroup of $O(d)$ that also commutes with a diagonal RMSNorm gain (appendix A), so they are covered exactly, not approximately—which is why our account connects to the permutation and linear-mode-connectivity literature [2, 7]. (iii) *Model merging*. Merging two independently trained models is, in these terms, an attempt to average two points that lie in *different* representatives of *different* classes; it succeeds only after the representatives are brought into a common frame, which is why permutation-alignment before averaging [2] helps for exactly the reason proposition 4.1 predicts. Our framework does not claim to solve merging, but it locates the difficulty precisely: without alignment, the average of two representatives need not lie in either class, so the merged function is not controlled by either teacher. This is the same coordinate sensitivity that governs grafting (section 6); merging is grafting’s symmetric, two-sided case.

A direction the theory suggests: reusing capability by alignment, not retraining. The same geometry points to a way of *amortizing* the cost of building capable models, which we flag as a consequence the framework predicts rather than one we test. Training general capability from scratch is expensive; but if that capability already resides in some donor block, proposition 3.2 says it is carried by the block’s output behavior, a class invariant—the block holds the capability, merely in its own representative (coordinate frame). Transplanting it into a new host should then require not retraining the block but only moving it into the host’s frame: estimate the low-capacity alignment $\mathbf{W}$ (definition 4.1, remark 4.1), graft in the aligned coordinates, and fine-tune only the seam. The heavy, capability-bearing weights are reused verbatim; only the light coordinate converter $\mathbf{W}$ is learned. prediction 6.3 is exactly this recipe read forward: alignment raises graft success most for the low-overlap pairs that arise when donor and host come from different models, so the models that would not stitch naively become stitchable once aligned. If it holds beyond the single-family setting we tested, the payoff is a training-time saving—capability assembled from pretrained modules rather than relearned.

Three cautions from our own results bound the promise, and none is incidental. First, $\mathbf{W}$ must stay *low capacity* (remark 4.1): a high-capacity bridge manufactures structure the donor lacks, at which point one is no longer reusing capability but retraining it under another name—the training-time argument collapses with it. Second, overlap sets a ceiling but depth governs sensitivity (section 6): grafts near the readout fail even at $\text{CKA} \approx 1$, so alignment does not make every transplant succeed. Third, this is *replication, not creation* ($S \preceq T$): assembly reuses capability that already exists in some donor, and originates none—so it shortens the path to a model *as capable as its parts*, not past them. Within those limits, the framework recasts “train a capable model” as “align and reuse capable parts,” and predicts when that substitution is available: exactly when boundary subspaces overlap or can be aligned into overlap. We test the overlap-predicts-success half of this (section 6); the cross-model assembly it motivates is left, with prediction 6.2, to future work.

A preliminary cross-width probe (section 7.8) qualifies the recipe: when the goal is *capability* rather than coordinate reuse, estimating $\mathbf{W}$ buys nothing over logit-KD alone (the two match across seeds while $\mathbf{W}$ doubles the cost). So $\mathbf{W}$ earns its keep only where *coordinates* must move—feature grafting, where a host reads the donor’s features directly—not where the output function alone is the target.

A consequence: collapsing multi-agent systems into one model. The same reading applies to a setting we do not test but that the framework organizes cleanly: distilling a multi-agent system—a debate, a planner with executors, an orchestrator over specialists—into a single model. Recent work trains one model to internalize multi-agent dynamics (e.g. debate consensus, critique-and-revision) so that explicit test-time interaction becomes an implicit capability of a single forward pass [1, 18]. In our terms, the multi-agent system is the teacher. Its *joint output function*—the answer the collective produces, with the reasoning that produced it—is the class invariant to target. Supervising the single model on that output function is exactly the output-function matching of sections 5 and 7, not a matching of the agents’ internal representations. The framework also predicts the boundary observed in that work: a single model can *replicate* the collective’s output function on the trajectories that cover it, but does not thereby *exceed* the collective ($S \preceq T$), and gains nothing on inputs the interaction trajectories do not reach—the multi-agent analogue of corpus-bounded restoration (section 7). Whether collapsing the interaction into the weights forfeits capability that genuinely *requires* test-time interaction (rather than merely expressing it) is, by our replication/creation distinction, the right question to ask—and one our controlled studies do not settle.

A representational reading of variance collapse. The same equivalence-class lens connects to a phenomenon usually described at the reward level. Iterated self-training (generate, verify, filter, retrain) can reduce the effective dimension of the student’s representation; with the participation ratio of the Gram spectrum $\{\mu_i\}$, $PR = (\sum_i \mu_i)^2 / \sum_i \mu_i^2$, a basis-invariant effective rank, a monotone decrease of PR across rounds is the representational analogue of reward-variance collapse ($\hat{\sigma}_R \equiv 0$): vanishing output diversity coincides with a shrinking feature dictionary, in the sense of representational capacity studied for superposition [6]. Because verifier reward is binary the signal is sparse and collapse can be fast; STaR [23] and ReST [10] mitigate it with temperature sampling and periodic re-anchoring. Relational objectives (family (A)) are a candidate further mitigation, since they constrain the Gram spectrum directly rather than letting it collapse onto a single passing mode. We flag this as a connection, not a result: it indicates that the representational and reward-level views of collapse are two readings of the same loss of class volume.

Practical implications: how should one distill? The geometry yields a short decision rule, organized by what one wants to transfer. *To transfer capability*, use logit (output-function) matching: it is the only objective natively defined on the joint quotient, so it targets the invariant capability depends on without any alignment step (proposition 3.2, Result 2). *To transfer representational geometry* when student and teacher may differ in width, use a relational loss (L_{rel} , L_{CKA}): it is basis-invariant, needs no alignment, and applies across dimensions—but, on its own, it does not restore capability, so it should accompany rather than replace the logit term. *To transfer a module by grafting*, where a downstream block will read features coordinatewise, first estimate a low-capacity alignment (section 4) and match in the aligned frame; expect success to track boundary subspace overlap (section 6). The combined objective $\mathcal{L} = L_{\text{logit}} + \lambda L_{\text{rel}}$ (eq. (14)) is the default: logits carry capability, the relational term carries structure, and neither is asked to do the other’s job. The single mistake the theory rules out is matching absolute features—raw MSE or cosine—which targets a coordinate system that does not exist.

9 Conclusion

The intuition that distillation happens “in the basis” is right in spirit and wrong in letter. A representation is not a feature matrix but an orthogonal–scaling equivalence class; absolute feature matching penalizes function–preserving changes within that class (proposition 2.1, table 1), and the network’s output function is precisely the invariant of the class that capability depends on (proposition 3.1). The experiments make the consequence concrete: aligning representations and restoring capability are near–orthogonal, and what transfers is the output–function invariant on the subspace the corpus covers—enough to rebuild a decoder from random initialization. The same invariants predict module–graft success through subspace overlap and recast the empirically decisive role of domain alignment as mediation through representational overlap.

Knowledge distillation should therefore be viewed not as feature matching, but as *supervision over representation equivalence classes*. Capability is transferred through their output–function invariants. What a teacher knows lives in the class; what transfers is the part of that class the objective and the corpus actually target.

If there is one sentence to carry away, it is this: *teacher supervision should target the class, not the coordinates*. The coordinates are an accident of training; capability is fixed by the one invariant the class determines—its output function—and distillation works when, and only when, it speaks in that invariant. We have argued the point geometrically, isolated it experimentally, and traced it through restoration, reconstruction, and grafting. Its value is as a lens: once a representation is seen as a class rather than a matrix, “what should we match?” answers itself—match what is defined on the quotient. The long list of distillation losses then collapses into one distinction: whether an objective is a function of the class, or of the basis we were never entitled to read.

References

- [1] Y. Luo, Y. Jin, W. Yu, M. Zhang, S. Kumar, X. Li, W. Xu, X. Chen, and J. Wang. AgentArk: Distilling multi-agent intelligence into a single LLM agent. *arXiv:2602.03955*, 2026.
- [2] S. K. Ainsworth, J. Hayase, and S. Srinivasa. Git Re-Basin: Merging models modulo permutation symmetries. In *International Conference on Learning Representations (ICLR)*, 2023.
- [3] Y. Bansal, P. Nakkiran, and B. Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems 34*, 2021.
- [4] C. Cortes, M. Mohri, and A. Rostamizadeh. Algorithms for learning kernels based on centered alignment. *Journal of Machine Learning Research*, 13:795–828, 2012.
- [5] N. Cristianini, J. Shawe-Taylor, A. Elisseeff, and J. Kandola. On kernel-target alignment. In *Advances in Neural Information Processing Systems 14*, 2001.
- [6] N. Elhage, T. Hume, C. Olsson, et al. Toy models of superposition. Anthropic technical report, *Transformer Circuits Thread*, 2022.
- [7] R. Entezari, H. Sedghi, O. Saukh, and B. Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations (ICLR)*, 2022.
- [8] A. Grattafiori et al. (Llama Team). The Llama 3 herd of models. *arXiv:2407.21783*, 2024.

- [9] A. Gretton, O. Bousquet, A. Smola, and B. Schölkopf. Measuring statistical dependence with Hilbert–Schmidt norms. In *Algorithmic Learning Theory*, pp. 63–77, 2005.
- [10] C. Gulcehre, T. Le Paine, S. Srinivasan, et al. Reinforced self-training (ReST) for language modeling. *arXiv:2308.08998*, 2023.
- [11] G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv:1503.02531*, 2015.
- [12] M. Klabunde, T. Schumacher, M. Strohmaier, and F. Lemmerich. Similarity of neural network models: A survey of functional and representational measures. *arXiv:2305.06329*, 2023.
- [13] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, PMLR 97:3519–3529, 2019.
- [14] N. Kriegeskorte, M. Mur, and P. Bandettini. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4, 2008.
- [15] J. M. Lee. *Introduction to Topological Manifolds*, 2nd edition. Graduate Texts in Mathematics, vol. 202. Springer, 2011.
- [16] K. Lenc and A. Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 991–999, 2015.
- [17] W. Park, D. Kim, Y. Lu, and M. Cho. Relational knowledge distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3967–3976, 2019.
- [18] B. S. Bijoy, M. S. Hasan, P. Alipoormolabashi, A. Sil, A. Balasubramanian, and N. Balasubramanian. ProST: Progressive sub-task training for Pareto-optimal multi-agent systems using small language models. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL)*, pp. 3357–3375, 2025.
- [19] Qwen Team. Qwen2.5 technical report. *arXiv:2412.15115*, 2024.
- [20] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio. FitNets: Hints for thin deep nets. In *International Conference on Learning Representations (ICLR)*, 2015.
- [21] P. H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.
- [22] F. Tung and G. Mori. Similarity-preserving knowledge distillation. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 1365–1374, 2019.
- [23] E. Zelikman, Y. Wu, J. Mu, and N. D. Goodman. STaR: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems 35*, 2022.

A Exactness of the assumption and the status of the propositions

When the absorbing identity is exact. Assumption 2.1 holds *exactly* for a $\mathbf{Q}$ when the operator $\mathbf{W}_{\text{in}}$ consuming $\mathbf{H}$ is linear and the intervening normalization commutes with $\mathbf{Q}$. Plain RMSNorm (norm rescaling, no learned gain) commutes with any $\mathbf{Q} \in \text{O}(d)$; practical RMSNorm carries a learned per-channel scale γ , which does not, since $\mathbf{Q}$ mixes channels while diagonal γ acts per axis. Exact commutation then holds only on the subgroup preserving γ (permutations, sign flips, rotations within equal- γ eigenspaces); for general $\mathbf{Q}$ the relation is *approximate*, with error set by the spread of γ .

Are the propositions exact theorems or approximations? The answer is precise once the group is named. On the subgroup $\mathcal{G}_0 \subseteq \text{O}(d) \times \mathbb{R}_+$ for which assumption 2.1 holds exactly—all of $\text{O}(d) \times \mathbb{R}_+$ for a linear reader with plain RMSNorm, and at least the γ -preserving subgroup for practical RMSNorm—the propositions propositions 2.1, 3.1 and 3.2 are *exact theorems*: the absorbing identity (2) is an algebraic equality, so $f^g = f$ holds with no error for every $g \in \mathcal{G}_0$, and the factorization $f = \bar{f} \circ \pi_{\text{joint}}$ over $\mathcal{G}_0$ is exact. What is *approximate* is only the *size* of the group: for a general $\mathbf{Q}$ not preserving γ , absorption incurs an error controlled by the spread of γ , so the equivalence class is at least $\mathcal{G}_0$ but the full $\text{O}(d) \times \mathbb{R}_+$ only approximately. This is why the theory is stated as exact propositions under assumption 2.1 while the experiments (section 7) report the consequences on real transformers, where the operative group is $\mathcal{G}_0$ enlarged by whatever near- γ -preserving directions the trained model tolerates: the qualitative claims (ill-posedness, output-only capability transfer) hold as soon as $\mathcal{G}_0$ is nontrivial, which it always is. Appendix F verifies the exact case on a normalization-free MLP.

B The maps in a transformer

Concretely, take $h = \mathbf{H}_\ell$ to be the residual stream after block ℓ (width d). The next block consumes it through linear projections: in attention, the query/key/value maps $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times m}$ applied to (normalized) h ; here $\mathbf{W}_{\text{in}}$ is any one of these and m the head or value width. The remainder of the block and all later blocks, the final norm, and the unembedding $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times V}$ compose into ρ , so $\phi_{\ell:L} = \rho \circ \psi$ maps the d -dimensional representation to the V -dimensional logits. A rotation $h \mapsto h\mathbf{Q}$ of the residual stream is absorbed by replacing each consuming projection $\mathbf{W}_{\text{in}} \mapsto \mathbf{Q}^\top \mathbf{W}_{\text{in}}$ (equivalently $\mathbf{W}_Q \mapsto \mathbf{Q}^\top \mathbf{W}_Q$, and so on): the products $(h\mathbf{Q})(\mathbf{Q}^\top \mathbf{W}_{\text{in}}) = h\mathbf{W}_{\text{in}}$ that enter every downstream computation are unchanged, because $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$. The only obstruction is the normalization between h and these projections. The simplest instance removes even that obstruction: in a plain MLP $x \rightarrow \mathbf{W}_1 \rightarrow \text{ReLU} \rightarrow h \rightarrow \mathbf{W}_2 \rightarrow \text{logits}$, the consuming operator is $\mathbf{W}_{\text{in}} = \mathbf{W}_2$ with $\tilde{\psi} = \text{id}$ (no normalization), so the cancellation (2a) is the exact linear identity $(h\mathbf{Q})(\mathbf{Q}^\top \mathbf{W}_2) = h\mathbf{W}_2$. Appendix F verifies this and the consequence (2b) to machine precision ($\sim 10^{-14}$), step by step, and shows that the un-compensated rotation changes the function.

C What a released model fixes: θ , not $\mathbf{H}$

Publishing a model releases its parameters θ (embedding, per-layer weights $\mathbf{W}_{\text{in}}^{(\ell)}$, unembedding), *not* a representation $\mathbf{H}$. The representation of definition 2.1 is a *derivative*, materialized only by running θ on a chosen probe set $\mathcal{X}$; two downloaders with different probes obtain different $\mathbf{H}$ from the same file. What is shipped is the ability to compute f ; what a student extracts is set by the probes it chooses (the “teacher $\times \mathcal{X}$ ” unit of definition 5.1).

Although $\mathbf{H}$ is identifiable only up to its class $[\mathbf{H}]$ (definition 2.2), the shipped θ nonetheless *fixes a representative*: it wires a specific $\mathbf{W}_{\text{in}}^{(\ell)}$ to read $\mathbf{H}$ in one coordinate frame, committing to a single point of the joint quotient (proposition 3.2) with no record that the counter-rotated $(c\mathbf{H}\mathbf{Q}, \theta^{\mathbf{Q},c})$ computes the same f . Function and file are consistent at two levels: the function depends only on the class, while the file commits to one frame. This is exactly why coordinate-sensitive operations across independently released models (grafting, section 6; an untied head, section 7.6) need an explicit alignment (definition 4.1). Each model was published in its own frame; there is no shared basis until one is estimated.

D Non-isotropic rescaling and the full symmetry group

A per-feature diagonal $\mathbf{\Lambda}$ can also be function-preserving when the next operator absorbs $\mathbf{\Lambda}^{-1}$, so the *full* symmetry group of the network is larger than $O(d) \times \mathbb{R}_+$. We do not claim our invariants capture that larger group; we claim only that absolute coordinate matching is ill-posed because the function is invariant to at least the $O(d)$ action (proposition 2.1), and that the admissible targets of section 3 are exactly the invariants of $O(d) \times \mathbb{R}_+$. Matching the diagonal part requires the alignment maps of section 4, not the relational invariants.

E Two objections to calling $\mathbf{H}$ the “representation”

The main text fixes our usage— $\mathbf{H}$ (activations) is the representation, $\mathbf{W}$ the reader. Two readers will each expect the word to mean something else; we address both.

The representation-theory reader expects a homomorphism $\varrho : \mathcal{G} \rightarrow GL(V)$ realizing an abstract group by linear maps. That is not what $\mathbf{H}$ is. A group does appear— $\mathcal{G} = O(d) \times \mathbb{R}_+$ —but it *acts on* the feature matrix $\mathbf{H}$ by $h \mapsto ch\mathbf{Q}$; $\mathbf{H}$ does not *represent* $\mathcal{G}$. In representation-theory terms the relevant object is the $\mathcal{G}$ -action and its orbits, not a representation of $\mathcal{G}$ on a vector space; $\mathbf{H}$ carries no such homomorphism, it is data. The “representation equivalence class” means the orbit of $\mathbf{H}$ under this action (definition 2.2), with nothing about characters, irreducibles, or intertwiners intended.

The computer-science reader may expect the opposite assignment: that the *weights* $\mathbf{W}$ —the fixed, shipped, linear-operator part of the model—are its “representation,” while the activations $\mathbf{H}$ are mere intermediate values. That intuition is reasonable ($\mathbf{W}$ is what one publishes, appendix C; a linear map is closer to the representation-theory sense). We use the opposite convention deliberately and for one reason: it is the established usage in representation learning, where “learned representation,” “hidden representation,” and the entire representation-similarity literature (CKA and its relatives, section 3) name the layer activations $\mathbf{H}$, not the weights. Our results are stated in that vocabulary so they connect to it.

F The claims on a plain MLP, where assumption 2.1 is exact

The paper’s account is not specific to transformers; it is a property of representation learning. To make this concrete—and to remove the one approximation the main text carries (RMSNorm’s learned gain does not commute with a general rotation, appendix A)—we reproduce the four core claims on a 2-layer MLP with *no normalization*, where assumption 2.1 holds *exactly*: a hidden rotation is absorbed by the next layer’s weights to machine precision. The network is $x \rightarrow \mathbf{W}_1 \rightarrow \text{ReLU} \rightarrow h \rightarrow \mathbf{W}_2 \rightarrow \text{logits}$, trained on a two-moons task (~ 0.998 train accuracy); $h \in \mathbb{R}^{N \times 16}$ is the

representation. Forward and backward passes are hand-written in NumPy (no autograd), so every step is inspectable. Table 8 collects the results.

Table 8: The four claims on a normalization-free MLP. assumption 2.1 is exact here, so the invariances hold to machine precision.

Claim	Result on the MLP
(1) proposition 2.1: absolute matching ill-posed	A function-preserving rotation $h \mapsto hQ$ sends $\ h - hQ\ _F^2 / N$ from 0 to 28.9 across $t \in [0, 1]$, while $1 - \text{CKA}$ and L_{rel} stay at 0 (to $< 10^{-9}$) for every t .
(2) assumption 2.1/proposition 3.1: joint invariance	The cancellation (2a) is exact here ($\tilde{\psi} = \text{id}$, no normalization): $\ (hQ)(Q^\top W_2) - hW_2\ \approx 10^{-14}$. Hence its consequence (2b)—the joint reparametrization ($h \mapsto hQ$, $W_2 \mapsto Q^\top W_2$)—preserves the function: $\max f^Q - f \approx 10^{-14}$. Rotating h <i>without</i> compensating W_2 —call this network f_{rot} , i.e. $h \mapsto hQ$ with W_2 left unchanged—changes it: $\max f_{\text{rot}} - f = 25.5$. Invariance is joint, not of h alone (remark 3.1).
(3) Representation $\neq$ function (the ablation)	A student hidden layer trained to match the teacher representation by a basis-invariant objective reaches $\text{CKA} = 0.79$ but $\text{KL}(T\ S) = 6.18$, top-1 agreement 0.50. Trained instead to match the output function (logits), it reaches $\text{KL} = 7 \times 10^{-4}$, top-1 ≈ 1.00 . Aligning geometry and restoring function are near-orthogonal, as in table 3.
(4) Procrustes alignment	For $H_r = HQ$, solving $Q_P = \arg \min_{Q'} \ H - H_r Q'\ _F$ reduces the relative residual from 1.41 to $\approx 10^{-14}$ and recovers $Q_P = Q^\top$ to $\approx 10^{-12}$: alignment finds the rotation, after which coordinate matching is well-posed.

The contrast with the main experiments is the point. In a transformer, assumption 2.1 is exact only on the γ -preserving subgroup and approximate otherwise, yet the same phenomena appear (tables 1 and 3). In the MLP there is no γ , the assumption is exact, and the invariances hold to machine precision. That the qualitative story is identical in both—feature matching penalizes a function-preserving change the invariants ignore (1), the output is invariant only under the joint action (2), aligning the representation does not restore the function (3), and an explicit alignment makes coordinate matching well-posed (4)—is evidence that the account is structural, not an artifact of any particular architecture. The script (NumPy only, no autograd) runs in seconds and reproduces these numbers to the reported precision across independent machines.

G Orbit, fiber, and quotient as runnable code

The three geometric objects the paper is built on—*orbit*, *fiber*, and *quotient*—are, for a reader who thinks in programs, three short pieces of code. This appendix gives a self-contained NumPy script (seed-fixed, no autograd) that constructs each one and prints what it does; remark 3.2 gives the same correspondences in prose.

We build them in the order a computer scientist finds easiest—starting from the two notions already familiar and ending at the two that are not. (i) *A feature is an IR*. The vector $\mathbf{h}$ below is an internal representation, the network’s analogue of a compiler’s SSA form: a scratch encoding the computation uses, not the answer it returns. (ii) *The invariant is the output*. Rewrite that IR—rotate $\mathbf{h}$, and update the reader to match—and the returned value is unchanged; the output

function is what stays fixed, exactly as an executable is unchanged by renaming a program’s local variables. These two steps need no group theory. (iii) *The set of IR-rewrites sharing one output is an orbit*, and (iv) *collapsing each orbit to a single object is a quotient*—and here the payoff is a distinction code makes unavoidable: there are *two* quotients, a coarser one that sees only the IR’s geometry (what CKA measures) and a finer one that sees the output function, and capability lives only on the finer one. Everything below runs from one setup: a single hidden representation h and the reader W_2 that consumes it, exactly as in the introduction’s three-line puzzle.

```
import numpy as np, hashlib
rng = np.random.default_rng(0)
N, d, V = 6, 8, 4
x = rng.standard_normal((N, d)); W1 = rng.standard_normal((d, d))
W2 = rng.standard_normal((d, V))
h = x @ W1                                # a hidden representation (N x d)
y = h @ W2                                # the output it produces (N x V)

def rotation(n):                            # a random orthogonal change of basis
    Q, _ = np.linalg.qr(rng.standard_normal((n, n))); return Q
def clone(h, W2, Q, c):                    # the joint rewrite: absorb (Q, c)
    return c * (h @ Q), (1.0 / c) * (Q.T @ W2)
def out_hash(h, W2):                       # a 'behavior' = hash of the output function
    return hashlib.sha256(np.round(h @ W2, 6).tobytes()).hexdigest()[:12]
def gram_id(h):                            # canonical id of the REPRESENTATION quotient
    G = h @ h.T                            # h h^T is invariant to h -> c h Q
    return hashlib.sha256(np.round(G/np.linalg.norm(G), 6).tobytes()).hexdigest()[:12]
```

Feature = IR; invariant = output. The starting point, before any orbit or quotient: the feature h is the network’s *intermediate representation*, and rewriting it—rotating and rescaling, with the reader updated in compensation—is a semantics-preserving transformation. The IR changes substantially ($\|h_{\text{ir}} - h\|$ large) while the output is bit-for-bit the same (identical hash), the way renaming an SSA temporary leaves the emitted executable unchanged. The output is the invariant; the feature is not.

```
r = np.random.default_rng(12345)          # isolated: doesn't perturb below
Q0, _ = np.linalg.qr(r.standard_normal((d, d))); c0 = 1.7
h_ir, W2_ir = clone(h, W2, Q0, c0)        # re-coordinatize the IR
print(f"IR changed? ||h_ir - h|| = {np.linalg.norm(h_ir - h):.3e}")
print(f"output same? {np.allclose(h_ir @ W2_ir, y)} "
      f"(hash {out_hash(h, W2)} -> {out_hash(h_ir, W2_ir)})")
```

```
IR changed? ||h_ir - h|| = 3.375e+01
output same? True (hash 4f39011d0b0d -> 4f39011d0b0d)
```

Orbit = all implementations of one program. An orbit is the set of networks obtained from (h, W_2) by the joint rewrite $(h, W_2) \mapsto (chQ, c^{-1}Q^T W_2)$. Each stores wildly different numbers in its hidden layer (large $\|h_{\text{var}} - h\|$) yet computes the identical output (identical hash), which is the orbit’s defining property.

```
print(f"{'variant':<20}{||h_var - h||:>15}{ 'output hash':>15}{ '==y?':>7}")
print(f"{'original':<20}{0.0:>15.3e}{out_hash(h,W2):>15}{ 'yes':>7}")
for k in range(1, 4):
    Q, c = rotation(d), float(rng.uniform(0.5, 2.0))
    hk, W2k = clone(h, W2, Q, c)
```

```
print(f"{'clone %d'%k:<20}-{np.linalg.norm(hk-h):>15.3e}"
      f"{'out_hash(hk,W2k):>15}-{('yes' if np.allclose(hk@W2k,y) else 'NO'):>7}")
```

variant	h_var - h	output hash	==y?
original	0.000e+00	4f39011d0b0d	yes
clone 1	2.618e+01	4f39011d0b0d	yes
clone 2	3.017e+01	4f39011d0b0d	yes
clone 3	1.963e+01	4f39011d0b0d	yes

Fiber = everything that maps to one output. Fixing a behavior (here, the output-hash of y), its *fiber* is the preimage under the output map—every $(h, \mathbf{W}_2)$ producing that behavior. All 200 joint rewrites land in it, because the rewrite is exactly the function-preserving move; a bare rotation that does *not* compensate the reader leaves the fiber. This is the code form of “same executable / same hash / same API”: the fiber is the set of sources that build to one artifact.

```
inside = sum(out_hash(*clone(h, W2, rotation(d), float(rng.uniform(0.5,2.0))))
            == out_hash(h, W2) for _ in range(200))
print(f"joint rewrites landing in fiber of y : {inside}/200")
y_rot = (h @ rotation(d)) @ W2      # rotate h but DON'T fix the reader
print("rotation without compensating reader :",
      "in" if np.allclose(y_rot, y) else "OUTSIDE", "the fiber")
```

```
joint rewrites landing in fiber of y : 200/200
rotation without compensating reader : OUTSIDE the fiber
```

Quotient = collapse each orbit to one point—but which quotient? This is the paper’s central distinction (proposition 3.2, fig. 2) made executable. Take *two* programs, A and B , that share the same hidden representation h but have *different* readers ($\mathbf{W}_2$ vs. a fresh $\mathbf{W}_2^B$), and make 30 clones of each. Collapsing by the representation quotient—keyed on the Gram invariant $h h^\top$, which is what CKA sees—merges A and B into *one* class: geometry alone cannot tell them apart. Collapsing by the joint quotient—keyed on the output function—keeps them as *two* pure classes. Capability lives on the finer, joint quotient; the representation quotient is strictly coarser.

```
W2b = rng.standard_normal((d, V))          # program B: different reader
rows = []
for name, ww in (("A", W2), ("B", W2b)):
    for _ in range(30):
        Q, c = rotation(d, float(rng.uniform(0.5, 2.0)))
        rows.append((name, *clone(h, ww, Q, c)))
joint, repq = {}, {}
for name, hk, wk in rows:
    joint.setdefault(out_hash(hk, wk), set()).add(name)
    repq.setdefault(gram_id(hk), set()).add(name)
print(f"representation quotient (gram) -> {len(repq)} class ; "
      f"A and B collapsed together: {any(len(v) > 1 for v in repq.values())}")
print(f"joint quotient (output func) -> {len(joint)} classes; "
      f"every class pure (one program): {all(len(v) == 1 for v in joint.values())}")
```

```
representation quotient (gram) -> 1 class ; A and B collapsed together: True
joint quotient (output func) -> 2 classes; every class pure (one program): True
```

The last block is fig. 2 in code: A and B sit in the same representation-quotient point (the same “meeting room,” remark 3.1) yet different joint-quotient points (different “seats”), and it is the seat—the output function—that capability reads. A distillation objective defined on the Gram invariant cannot distinguish A from B ; one defined on the output function can. That is the paper’s thesis, executable in forty lines. The full script—printing every intermediate result—is distributed with the paper’s code release.